THE SIZE RAMSEY NUMBER OF A PAIR OF STAR FORESTS: A PROOF OF THE BURR–ERDŐS–FAUDREE–ROUSSEAU–SCHELP FORMULA (ERDŐS PROBLEM #561)

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Abstract. We prove the diagonal-sum upper bound for the size Ramsey number of two arbitrary star forests and develop the proposed matching-certificate proof of the corresponding equality. The lower-bound argument reduces to five classes of minimal failures under arbitrary subgraph deletion. The degree exclusions use exact certificate costs to force successive maximum matchings on one retained graph, followed by restoration of every omitted edge. Several foundational proofs remain explicitly deferred, so the claimed equality in this draft depends on those technical inputs. The main result of this paper was obtained with the assistance of the Ansatz system, an automated proof-search and verification system.

1. Introduction

The problem. For finite simple graphs G_1, G_2, write $H \to (G_1, G_2)$ if every red–blue coloring of $E(H)$ contains a red copy of G_1 or a blue copy of G_2. Copies are ordinary subgraphs. The size Ramsey number $\widehat{R}(G_1, G_2)$ is the least edge count of such a graph H. We write $\widehat{R}(G) = \widehat{R}(G, G)$. The star-forest problem asks for this number when both targets are disjoint unions of nonempty stars. The formula was conjectured in [BEFRS78, p. 194], where the uniform case (all stars within each forest of the same size) was proved; the conjecture and the previously known cases are surveyed in [DJKR21, Section 1].

For the original conjecture, the bounded-degree colouring paper, the uniform-star-forest preprint, and the problem database, see [BEFRS78, p. 194], [GyScO2], [FLN26], and [EP561], respectively.

The formula. The component sizes may repeat and may equal one.

Theorem 1.1 (Star-forest formula)**.** *Let $s, t \geq 1$, and let*

$$n_1 \geq \cdots \geq n_s \geq 1, \qquad m_1 \geq \cdots \geq m_t \geq 1$$

be integer lists. Let

$$F_1 = \bigsqcup_{i=1}^{s} K_{1,n_i}, \qquad F_2 = \bigsqcup_{j=1}^{t} K_{1,m_j},$$

where $K_{1,h}$ has one center, h leaves, and its h center–leaf edges. For $2 \leq k \leq s+t$, put

$$\ell_k = \max_{\substack{1 \leq i \leq s,\ 1 \leq j \leq t \\ i+j=k}} (n_i + m_j - 1).$$

Then

$$\widehat{R}(F_1, F_2) = \sum_{k=2}^{s+t} \ell_k.$$

The theorem contains all previously established cases: the uniform case [BEFRS78], the dominance case of [GyScO2] (restated as [DJKR21, Theorem 1.3]), the case of one star against an arbitrary star forest [DJKR21, Theorem 2.3], and the further special cases of [DJKR21, Section 2]. These are consequences obtained by specializing the lists.

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[prime/blocker] Theorem 1.1 is retained as the supplied target assertion. Its proof in this manuscript depends on the explicitly deferred technical inputs below. Those inputs have neither complete proofs here nor a verified companion bibliography entry. This draft must not be described as a complete independent proof.

[prime/blocker] The supplied material includes assertions inherited from entire earlier collections of statements whose contents were not provided. It also includes extensive intermediate numerical restrictions, physical exchange statements, and frame-activity inventories. The requested instruction to omit those inventories conflicts with preservation of every supplied mathematical assertion. This manuscript develops the final exclusion route; it does not preserve that additional material unabridged. In particular, the omitted inherited statements cannot be reconstructed from their names or dependency references.

Status of the proof. The proof is computer-assisted, with the machine-verified certificate [Rec] deposited with this paper. All statements in this paper and in [Rec] were produced and verified within the Ansatz system, an automated proof-search and verification system. Every statement was additionally reviewed by two independent automated audits. None of these automated checks constitutes checking by a human referee. This paper proves the formula from twenty-two explicitly stated inputs, each of which is a theorem of [Rec]. These inputs are Propositions 2.2 and 4.3, Theorem 4.4, and Propositions 4.6, 5.1, 5.2, 5.3, 5.4, 5.5, 6.4, 6.5, 6.8, 6.9, 6.12, 6.13, 6.15, 6.19, 6.23, 6.24, 6.25, 6.26, and 6.27. Readers who wish to examine the inputs should begin with the reduction, developed in [I].

Method and organization. A certificate partitions the edges into two parts of bounded maximum degree and two parts of bounded matching number. Its cost requires feasibility for every integer allocation of the matching budget. The reduction in Section 4 shows that five classes $\mathcal{M}_4, \mathcal{M}_6, \mathcal{M}_8, \mathcal{M}_{10}, \mathcal{M}_{12}$ suffice for the lower bound. Each class uses minimality under every proper subgraph, including spanning edge deletions. The source profiles in Section 5 supply exact costs. Section 6 uses those costs to force compatible matchings and a final odd-cycle remainder. Every edge incident with an omitted vertex is then restored in an indexed star.

Section 2 fixes the certificate conventions. Section 3 proves the upper bound. Section 7 assembles the five exclusions, and Section 8 records the logical scope of the method.

Remark 1.2 (Disclosure and verification status). The main result of this paper was obtained with the assistance of the Ansatz system, an automated proof-search and verification system. The supplied mathematical development is reported to have undergone machine verification; independent expert review is pending. This manuscript has not independently reverified the omitted technical proofs. Because automated systems have limitations, it is possible that we have missed related references in the literature, and we welcome comments from experts.

[author/blocker] No author, affiliation, or email was supplied.

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2. Preliminaries

We use the matching terminology of [LP86]. All graphs are finite and simple. The matching number, independence number, maximum degree, edge count, and minimum vertex-cover cardinality are denoted by $\nu(G), \alpha(G), \Delta(G), e(G), \tau(G)$. The empty graph has matching number and maximum degree zero. An edge set is regarded as a spanning graph when one of these quantities is applied to it. Deleting a matching deletes its edges and retains all vertices. Deleting a vertex set means taking the induced graph on the remaining vertices.

Definition 2.1 (Certificates and costs)**.** Let A, B, x, y be nonnegative integers. An $(A, B; x, y)$-certificate of G is a partition

$$E(G) = Q_R \,\dot\cup\, Q_B \,\dot\cup\, P_R \,\dot\cup\, P_B$$

such that

$$\Delta(Q_R) \le A, \qquad \Delta(Q_B) \le B, \qquad \nu(P_R) \le x, \qquad \nu(P_B) \le y.$$

Let $c_{A,B}(G)$ be the least nonnegative integer T for which every nonnegative integer pair $x+y=T$ has such a certificate. Set

$$c_i(G) = \max_{\substack{A,B\ge 0\\ A+B=i}} c_{A,B}(G), \qquad c_0(G) = c_{0,0}(G), \qquad \Sigma(G) = \sum_{i\ge 0} c_i(G).$$

Thus c_i is also the quantity denoted c_i^* in formulations using row maxima. Define

$$\mathrm{ex}(G) = c_0(G) - \nu(G), \qquad \mathrm{ps}(G) = e(G) - \nu(G) - \sum_{i\ge 1} c_i(G), \qquad \mathrm{gap}(G) = \Sigma(G) - e(G).$$

We call G *LP-good* when $\Sigma(G) \le e(G)$. The gap is a signed integer.

Proposition 2.2 (Degree splitting with a covering matching)**.** *Let $A, B \ge 0$ be integers with $D = A+B \ge 1$. Let G satisfy $\Delta(G) \le D$. Assume that a matching covers every vertex of degree D. Then $E(G)$ has a red–blue coloring with red maximum degree at most A and blue maximum degree at most B.*

A complete proof is recorded in [Rec]; a human-readable development is given in [I].

Proposition 2.3 (Basic certificate identities)**.** *For every finite simple graph G the costs are well defined. They have the following properties.*

(a) *Every allocation of total T is certified at capacities (A,B) if and only if $T \ge c_{A,B}(G)$.*

(b) *The costs are symmetric in A, B and nonincreasing in each capacity. In particular $c_i(G) \le c_0(G)$.*

(c) *For a vertex-disjoint union,*

$$c_{A,B}(G_1 \sqcup G_2) \le c_{A,B}(G_1) + c_{A,B}(G_2), \qquad c_i(G_1 \sqcup G_2) \le c_i(G_1) + c_i(G_2).$$

(d) *Under the hypotheses of Proposition 2.2, $c_{A,B}(G) = 0$. Consequently $c_i(G) = 0$ for $i \ge \Delta(G)+1$.*

(e) *For every vertex v,*

$$c_{A,B}(G) \le c_{A,B}(G-v) + 1, \qquad c_i(G) \le c_i(G-v) + 1.$$

(f) *If G' is a subgraph of G, then $c_{A,B}(G') \le c_{A,B}(G)$.*

(g) *The zero cost satisfies*

$$\nu(G) \le c_0(G) \le \tau(G), \qquad 0 \le \mathrm{ex}(G) \le \tau(G) - \nu(G).$$

(h) *One has*

$$\sum_{i\ge 1} c_i(G) \le e(G) - \nu(G), \qquad \mathrm{ps}(G) \ge 0.$$

(i) *The exact identity*

$$\mathrm{gap}(G) = \mathrm{ex}(G) - \mathrm{ps}(G)$$

holds. Thus G is LP-good if and only if $\mathrm{ps}(G) \ge \mathrm{ex}(G)$, $\Sigma(G) \le e(G) + \mathrm{ex}(G)$, and every graph with $\tau(G) = \nu(G)$ is LP-good.

Proof. At total $2\nu(G)$ one budget is at least $\nu(G)$; put all edges in that matching-bounded part. This proves existence. If total T works, an allocation (x,y) of total $T+1$ dominates $(x-1,y)$ when $x>0$, and dominates $(0,T)$ when $x=0$. This proves (a). Exchanging the colors proves symmetry, and relaxing capacities proves (b).

For (c), write $T_i = c_{A,B}(G_i)$. Given $x+y = T_1+T_2$, take

$$x_1 = \min(x, T_1), \quad y_1 = T_1 - x_1, \quad x_2 = x - x_1, \quad y_2 = y - y_1.$$

All four numbers are nonnegative and $x_i + y_i = T_i$. Unite certificates part by part. Maximum degree takes the larger component value, and matching number is additive on vertex-disjoint unions. Taking maxima over the capacities gives the row inequality.

Proposition 2.2 gives (d) by taking both paid parts empty. When $i > \Delta(G)$, the empty matching covers the empty set of degree-i vertices.

For (e), use a certificate on $G-v$ and put all edges incident with v in a paid part whose budget was increased by one. A matching uses at most one of those edges. Restricting a certificate to a subgraph proves (f).

If $T < \nu(G)$, the allocation $(0, T)$ at capacities $(0, 0)$ fails, which proves the lower bound in (g). For the upper bound, partition a minimum vertex cover into sets of sizes x, y with $x + y = \tau(G)$. Assign to the first paid part every edge meeting the first set, and put the remaining edges in the second part. The two covering sets bound the respective matching numbers.

We prove (h) by induction on the number of vertices. The edgeless case follows from (d). Put $D = \Delta(G) \geq 1$. If some degree-D vertex v is missed by a maximum matching, then $\nu(G - v) = \nu(G)$, and (d), (e), and induction give

$$\sum_{i \geq 1} c_i(G) \leq \sum_{i=1}^{D} c_i(G - v) + D \leq e(G - v) - \nu(G - v) + D = e(G) - \nu(G).$$

Otherwise every maximum matching covers every degree-D vertex. Part (d) gives $c_D(G) = 0$. Choose any degree-D vertex v. Since $\nu(G - v) \geq \nu(G) - 1$,

$$\sum_{i \geq 1} c_i(G) \leq \sum_{i=1}^{D-1} c_i(G - v) + D - 1 \leq e(G) - D - \nu(G) + 1 + D - 1.$$

This proves (h). Finally, expanding the definitions gives (i). □

Definition 2.4 (Missing allocations and minimal failures)**.** If $c_0(G) = 0$, put $\operatorname{tr}(G) = 0$. Otherwise a first budget $x \in \{0, \ldots, c_0(G) - 1\}$ is *missing* when $E(G)$ has no partition into two parts of matching numbers at most x and $c_0(G) - 1 - x$. Let $\operatorname{tr}(G)$ be the largest length of a consecutive run of missing budgets, with value zero if there are none. Put

$$h_-(D) = \max\left(4, \lfloor D/2 \rfloor + 1\right), \qquad h_0(D) = \max\left(4, \lceil D/2 \rceil + 1\right).$$

The property T_- is

$$\operatorname{gap}(G) > 0 \implies \left[\operatorname{gap}(G) = 1 \text{ and } \operatorname{tr}(G) + \operatorname{ex}(G) \leq h_-(\Delta(G))\right].$$

The property T_0 uses h_0 instead.

A *closed d-lump* is a connected d-regular graph of odd order whose every vertex deletion has a perfect matching. It is *totally critical* when every vertex deletion strictly lowers all positive rows through d. For $d \in \{4, 6, 8, 10, 12\}$, let $\mathcal{M}_d$ consist of all graphs of maximum degree exactly d which fail T_- while every proper subgraph satisfies T_-. A proper subgraph may delete vertices, edges, or both. Let N_d denote the assertion $\mathcal{M}_d = \varnothing$.

Setup 2.5 (Target lists and suffixes)**.** For the reduction, write the target lengths as s_f, t_f to distinguish them from source parameters. Fix positive nonincreasing lists and their forests F_1, F_2, and put

$$M = s_f + t_f - 1, \qquad \ell_j = \max_{i+l=j}(n_i + m_l - 1), \qquad L = \sum_{j=2}^{M+1} \ell_j.$$

Every maximum uses valid original indices. For $0 \leq q < M$, put

$$L_q = \sum_{j=q+2}^{M+1} \ell_j, \qquad T_q = \bigsqcup_{j=2}^{q+1} K_{1,\ell_j}.$$

Let $\mathcal{A}_q$ contain every ordered pair of nonempty actual suffixes obtained by deleting α leading components of F_1 and β leading components of F_2, where $\alpha + \beta = q$, $0 \leq \alpha < s_f$, and $0 \leq \beta < t_f$.

Notation 2.6 (Source parameters)**.** For a graph $F \in \mathcal{M}_d$, use

$$k = \nu(F), \quad a = \alpha(F), \quad E = \operatorname{ex}(F), \quad \lambda = 2E - 3 - (k - a),$$
$$\delta = c_0(F) - c_2(F), \qquad \epsilon = c_0(F) - c_1(F), \qquad q_{\mathrm{def}} = \delta - \epsilon,$$
$$p = E - 2 - \lambda, \qquad r = k - \lambda - 1 + \delta, \qquad s = 2\lambda + 3 - 2\delta,$$
$$\beta = p + q_{\mathrm{def}}, \qquad u = r + s.$$

At degree eight, $q_8 = c_4(F) - r$ and $\eta = \lambda - \delta$. The quantities q_8 and q_{def} are distinct. An equality sector has $\delta = 0$; a strict sector has $\delta > 0$. All capacities, budgets, costs, cardinalities, phase lengths, and matching sizes are nonnegative integers. Differences are signed unless a stated result gives their sign.

Definition 2.7 (Native rows and processes)**.** For degrees six, eight, ten, and twelve, a *native row* is an induced subgraph A of F satisfying

$$|V(A)| = r, \quad \Delta(A) \le 2, \quad \mathrm{ps}(A) = 0, \quad \nu(A) = p + \delta, \quad c_1(A) = \beta, \quad c_2(A) = p.$$

A complete source process deletes a maximal independent set and then arbitrary current-maximum-degree vertices until the remaining maximum degree is at most two. A full degree-j phase consists of all deletions at current degree j. An actual prefix is an induced remainder after the initial independent-set deletion. A whole component is an entire connected component. Paths P_j, cycles C_j, and complete graphs K_j are indexed by vertex order. A graph is j-good when its maximum degree is at most j and one matching covers every degree-j vertex.

3. The upper bound

In this section we prove the upper bound using nested component thresholds.

Lemma 3.1 (Embedding by thresholds)**.** *Let $a_1 \ge \cdots \ge a_q \ge 1$, with $q \ge 1$. Let H be a vertex-disjoint union of $N \ge 0$ stars with positive edge counts, and let c_h be the number of edges of a fixed color in component h. Then H contains a copy of $\bigsqcup_{i=1}^{q} K_{1,a_i}$ in that color if and only if*

$$\big|\{h : c_h \ge a_i\}\big| \ge i \qquad (1 \le i \le q).$$

Proof. Every nonempty target star lies in one host component. Every edge in that component meets its designated center, so every embedded target star there contains that center, including a target $K_{1,1}$. Vertex-disjoint target components therefore occupy distinct host components. The first i targets require i distinct components with at least a_i edges of the chosen color.

Conversely, choose the host components in target order. At step i, at least i components meet the threshold a_i and at most $i-1$ have already been used. Choose an unused one and take exactly a_i of its edges in the chosen color. These edges form a star, and different steps use disjoint vertex sets. If $N = 0$, the first threshold fails and no nonempty target can occur. □

Proposition 3.2 (The diagonal star construction)**.** *For the lists in Theorem 1.1,*

$$H = \bigsqcup_{k=2}^{s+t} K_{1,\ell_k} \quad \textit{satisfies} \quad H \to (F_1, F_2), \qquad e(H) = \sum_{k=2}^{s+t} \ell_k.$$

Consequently $\widehat{R}(F_1, F_2) \le \sum_k \ell_k$.

Proof. For $2 \le k \le s+t$, choose $i = \max(1, k-t)$ and $j = k - i$. Then $1 \le i \le s$ and $1 \le j \le t$. Thus every diagonal set is nonempty, finite, and consists of positive integers. The construction has $s + t - 1$ positive-edge components and the stated edge count.

Fix a coloring, and let r_k, b_k be its red and blue edge counts on component k. Suppose neither target occurs. Lemma 3.1 gives indices i, j for which

$$A = \{k : r_k \ge n_i\}, \qquad B = \{k : b_k \ge m_j\}$$

have sizes at most $i-1, j-1$. For $2 \le k \le i+j$, take $v = \min(i, k-1)$ and $w = k - v$. Then $1 \le v \le i$, $1 \le w \le j$, and

$$\ell_k \ge n_v + m_w - 1 \ge n_i + m_j - 1.$$

If $k \notin A \cup B$, integrality gives $r_k + b_k \le n_i + m_j - 2$, a contradiction. Hence $\{2, \ldots, i+j\} \subseteq A \cup B$. Its size is $i + j - 1$, whereas $|A \cup B| \le i + j - 2$. This contradiction proves arrowing. The argument includes repeated sizes, unit stars, and either target length equal to one. □

4. The reduction

The goal of this section is to state the precise lower-bound reduction and prove its elementary deletion and replacement steps.

Lemma 4.1 (Actual suffixes after vertex deletion)**.** *Use Setup 2.5. Let $X = A \dot\cup B \subseteq V(G)$, with $|A| = \alpha < s_f$, $|B| = \beta < t_f$, and $q = \alpha + \beta \geq 1$. Every coloring of $G - X$ avoiding the corresponding actual suffix pair extends to a coloring of G avoiding F_1, F_2. Color edges within A and from A to $G - X$ red, and edges within B and from B to $G - X$ blue. Every edge between A and B may receive either color independently. Consequently $G \to (F_1, F_2)$ implies that $G - X$ arrows every member of $\mathcal{A}_q$.*

Proof. Every red edge incident with B has its other endpoint in A. A connected red subgraph containing an edge and meeting X must therefore contain a vertex of A. The corresponding blue assertion holds with B. Thus at most α components of a red target copy could meet X.

If $r \leq \alpha$ red components meet X, list the untouched component indices as $i_1 < \cdots < i_{s_f - r}$. For $1 \leq l \leq s_f - \alpha$, $i_l \leq l + r \leq l + \alpha$. Hence $n_{i_l} \geq n_{\alpha+l}$. Trimming the first $s_f - \alpha$ untouched stars gives the forbidden red suffix in $G - X$. The blue argument uses $j_l \leq l + \beta$ and $m_{j_l} \geq m_{\beta+l}$. This also covers copies meeting X through leaves, since no center assumption was used. □

Proposition 4.2 (The edge-minus-matching potential)**.** *Let $\mathcal{A}$ be a finite nonempty family of positive star-forest pairs, and put $N = \max_{\gamma \in \mathcal{A}}(s_\gamma + t_\gamma - 1)$. For $2 \leq j \leq N + 1$, let $\bar\ell_j$ be the largest $n_i^\gamma + m_l^\gamma - 1$ with valid indices $i + l = j$. If G arrows each pair separately, then*

$$e(G) - \nu(G) \geq \sum_{j=2}^{N+1} (\bar\ell_j - 1).$$

For one pair, every Ramsey host satisfies

$$e(G) \geq L + \nu(G) - M.$$

In particular, $\nu(G) \geq M$ implies $e(G) \geq L$. If $e(G) < L$, then

$$\nu(G) \leq M - 1, \qquad 1 \leq L - e(G) \leq M - \nu(G).$$

Proof. Write $\Phi(H) = e(H) - \nu(H) \geq 0$. If H has no coloring with degree capacities A, B, there is a maximum-degree vertex v with $\Phi(H) - \Phi(H - v) \geq A + B$. To see this, put $D = A + B$. When $\Delta(H) \geq D + 1$, any maximum-degree vertex works because vertex deletion lowers matching number by at most one. When $\Delta(H) = D \geq 1$, a matching covering all degree-D vertices would give the forbidden coloring by Proposition 2.2. Some maximum matching therefore misses a degree-D vertex; deleting it does not lower the matching number. The case $\Delta(H) < D$ would admit the coloring by the same proposition, using the empty covering matching. For $D = 0$, a graph with no such coloring has an edge, and $\Delta(H) - 1 \geq 0$ suffices.

A member of the family attaining N supplies valid index sums at every $j = 2, \ldots, N+1$, so all maxima are defined. After deleting $q < N$ vertices, choose a pair and indices attaining $\bar\ell_{q+2}$. Lemma 4.1, or the identity when $q = 0$, says that the remainder arrows their two nonempty suffixes. It has no coloring of degree capacities $n_i^\gamma - 1, m_l^\gamma - 1$. Delete a maximum-degree vertex paying at least $\bar\ell_{q+2} - 1$ in Φ. The remainder has an edge before this deletion, so the recursion is valid.

Summing the N losses and using nonnegativity of the final potential gives the family assertion. For one pair there are M diagonal entries, so the right side is $L - M$. Rearrangement and integrality give the remaining conclusions. □

Proposition 4.3 (Canonical pools and exact host normalization)**.** *Use Setup 2.5, and let H be disjoint from T_q. Then*

$$H \sqcup T_q \to (F_1, F_2) \quad \Longleftrightarrow \quad H \text{ arrows every member of } \mathcal{A}_q.$$

Every avoiding coloring of the union determines an avoided actual suffix pair without changing its colors on H. It can be replaced by an avoiding coloring with every pool star monochromatic. Conversely, every avoiding coloring of any member of $\mathcal{A}_q$ extends by monochromatic pool stars.

For $X \subseteq V(G)$ with $|X| = q < M$, put $G^\star_X = (G - X) \sqcup T_q$, $c_X = e(G) - e(G - X)$, and $h_X = \nu(G) - \nu(G - X)$. Then

$$e(G^\star_X) = e(G) - c_X + \sum_{j=2}^{q+1} \ell_j, \qquad \nu(G^\star_X) = \nu(G) - h_X + q.$$

Arrowing passes from G to $G^\star_X$, and avoidance pulls back to G while preserving every color on $G - X$.

Finally, assume $G \to (F_1, F_2)$, $e(G) - \nu(G) = L - M$, and $\nu(G) < M$. Fix a maximum matching Q. There exist $q < \nu(G)$ and $|X| = q$ such that, for $H = G - X$,

$$\nu(H) = \nu(G) - q, \qquad e(H) = e(G) - \sum_{j=2}^{q+1} \ell_j, \qquad \Delta(H) = \ell_{q+2} - 1 \geq 3.$$

The restriction of Q is maximum. The graph $G^\star_X$ has the same edge count, matching number, and potential as G. It is obtained by successively detaching full-degree vertices as the canonical prefix stars.

Proof. Scan the pool stars in order. Before star h, let α, β count the red and blue target heads already assigned; then $\alpha+\beta = h-1$. Avoidance prevents either target from being exhausted. The current capacity is at least $n_{\alpha+1} + m_{\beta+1} - 1$, so one color supplies its current head. Assign that head and continue. At the end, the unchanged core avoids the actual suffix pair indexed by the resulting α, β.

For the converse, prescribe any sequence of α red and β blue pool stars. At each step its capacity is at least the next head in its prescribed color. For one monochromatic host star of capacity at least the largest remaining target component, absence of the target in the union is equivalent to absence of the suffix in the core. A copy using the host star uses at most one target component there. The remaining core components contain the suffix after the largest component is deleted, by their nonincreasing sizes. A core suffix together with the host star gives the reverse implication. Applying this equivalence successively proves the assertion, including $q = 0$.

Lemma 4.1 proves arrowing after replacement. For a pullback, use the suffix selected by the pool scan, partition X into sets of sizes α, β, and apply that lemma. The pool has q matching edges and the displayed edge count, which proves the two identities.

For the last assertion, the required tight greedy-complement statement is: after every allowed greedy deletion, the surviving restriction Q_h is maximum and, for $J_h = (G - X_h) - Q_h$,

$$e(J_h) = \sum_{j=h+2}^{M+1} (\ell_j - 1), \qquad \Delta(J_h) = \ell_{h+2} - 1.$$

Delete a vertex of residual degree ℓ_{h+2} whenever one exists. It is Q_h-covered and has degree $\ell_{h+2} - 1$ in J_h. Each such deletion loses one matching edge and exactly ℓ_{h+2} edges. The process must stop before $\nu(G)$ deletions: otherwise the residual is edgeless but still arrows nonempty suffixes. At stopping, the maximum degree equals $\ell_{q+2} - 1$. A second required input states that a residual of maximum degree at most two, after an actually paid prefix of this kind, has at least L_q edges. It would give $e(G) \geq L$, so the terminal degree is at least three. The replacement identities now give the final assertions. □

A complete proof is recorded in [Rec]; a human-readable development is given in [I].

Theorem 4.4 (Contained minimal obstruction)**.** *Use Setup 2.5. If H arrows every entire prescribed-color member of $\mathcal{A}_q$ and $e(H) < L_q$, then an actual component C of H satisfies*

$$\mathrm{ex}(C) \geq M - q - \nu(H) \geq 4, \qquad \mathrm{gap}(C) \geq 1.$$

It contains a subgraph F which is a totally critical closed d-lump, where

$$d \in \{4, 6, 8, 10, 12\}, \qquad d \leq \Delta(H), \qquad \mathrm{gap}(F) = 1, \qquad \mathrm{tr}(F) + \mathrm{ex}(F) \geq h_-(d) + 1.$$

Every proper subgraph of F satisfies T_-, and every proper subgraph of excess at least four is LP-good. Containment need not be induced. The component excess bound is not asserted for F.

If $|V(F)| = 2k + 1$ and $\mathrm{ex}(F) \geq 4$, then for every edge e,

$$r_i(e) := c_i(F) - c_i(F - e) \in \{0, 1\}, \quad r_i(e) = 0 \ (i > d), \quad r_d(e) = 1,$$

and

$$\nu(F-e)=k,\quad \operatorname{ex}(F-e)=\operatorname{ex}(F)-r_0(e),\quad R(e):=\sum_{i=0}^{d} r_i(e)\ge 2,\quad \operatorname{gap}(F-e)=2-R(e)\le 0.$$

At degree four, every proper subgraph J satisfies

$$\operatorname{gap}(J)>0 \Longrightarrow [\operatorname{gap}(J)=1,\ \operatorname{tr}(J)+\operatorname{ex}(J)\le 3],$$

and every proper J with $\operatorname{ex}(J)\ge 3$ is LP-good.

A complete proof is recorded in [Rec]; a human-readable development is given in [I].

The finite-minimality step in this theorem is explicit. Once a component contains an induced T_- failure, choose a failing subgraph with the least edge count and then the least vertex count. Every proper subgraph has fewer edges or has the same edges and fewer vertices, so it satisfies T_-. A further induced failing subgraph of this chosen graph must equal it. This explains why the five-degree conclusion, when supplied, applies to the chosen graph and retains actual host containment.

Definition 4.5 (The rigid class)**.** The class $\mathcal{L}_{\mathrm{rig}}$ consists of closed d-lumps with $d\in\{4,6,8,10,12\}$, gap one, and every proper subgraph satisfying T_-. Put $k=\nu(F)$, $E=\operatorname{ex}(F)$, and $p_d=3$ for $d=4$, $p_d=4$ otherwise. Require

$$E\ge 4\ (d=4),\quad E\ge 5\ (d=6,8),\quad E\ge 6\ (d=10,12),\quad E\ne 7\ (d=12).$$

Require also:

(a) $c_i(F-v)=c_i(F)-1$ for every vertex and $0\le i\le d$.

(b) For every nonempty set S of edges incident with one vertex, the drops $r_i(S)=c_i(F)-c_i(F-S)$ belong to $\{0,1\}$, $r_d(S)=1$, and $\sum_{i<d} r_i(S)\ge |S|$.

(c) For every nonempty vertex set X, put $j=|X|$, $J=F-X$, and $b=j+\nu(J)-k$. If $b\le E-p_d$, then

$$c_d(J)=0,\qquad 1\le c_i(F)-c_i(J)\le j\quad (0\le i<d),$$

and, with $t_i=j-c_i(F)+c_i(J)$, $\sum_{i<d} t_i\le e(F[X])$.

Proposition 4.6 (Universal graph input)**.** *The following assertions are equivalent:*

(i) *Every finite simple graph satisfies T_-.*

(ii) *Every finite simple graph satisfies T_0.*

(iii) *Every totally critical closed d-lump of gap one, for $d\in\{4,6,8,10,12\}$, satisfies $\operatorname{tr}(F)+\operatorname{ex}(F)\le h_-(d)$.*

(iv) *$\mathcal{L}_{\mathrm{rig}}$ is empty.*

(v) *Every member of $\mathcal{L}_{\mathrm{rig}}$ satisfies that same bound.*

Under these assertions every C with $\operatorname{ex}(C)\ge h_0(\Delta(C))$ is LP-good. Every below-L_q common host in Theorem 4.4 contains a member of $\mathcal{L}_{\mathrm{rig}}$ of degree at most its maximum degree.

A complete proof is recorded in [Rec]; a human-readable development is given in [I]. In particular, the implication from universal T_- to emptiness uses the additional assertion that a graph of gap one all of whose induced subgraphs satisfy T_- has excess at most three. The converse uses existence and all deletion identities of a proper-subgraph-minimal failure.

Theorem 4.7 (The five exclusions suffice)**.** *Assume $N_4, N_6, N_8, N_{10}, N_{12}$. Then every finite simple graph satisfies T_- and T_0. For every q in Setup 2.5, every graph H with $e(H)<L_q$ has a coloring avoiding both entire targets of some member of $\mathcal{A}_q$. Every such coloring extends over the unchanged canonical pool T_q, and across every specified outer q-vertex set, to avoidance of the unchanged original targets while retaining all colors on H. Under these five hypotheses, Theorem 1.1 holds and its diagonal star construction attains the minimum.*

Proof. A member of $\mathcal{L}_{\mathrm{rig}}$ is, by the strengthened minimal-failure input, a T_- failure in one of the five degrees. It would belong to the corresponding $\mathcal{M}_d$. Thus $\mathcal{L}_{\mathrm{rig}}$ is empty, and Proposition 4.6 gives the two universal graph properties.

If a below-budget H arrowed every member of $\mathcal{A}_q$, Theorem 4.4 would give a member of one of the five $\mathcal{M}_d$, contradicting its emptiness. Negating all-family arrowing gives an actual avoided suffix pair and a coloring avoiding both its entire targets. Proposition 4.3 and Lemma 4.1 give the two extensions. At $q = 0$, the suffix family consists of the original pair and $L_0 = L$. Hence every graph with fewer than L edges avoids the pair. Proposition 3.2 supplies the matching upper bound. No converse from the Ramsey formula to the five exclusions is used. □

5. Universal source structure

In this section we state the source profiles needed to apply the exclusion mechanism to every member of each minimal-failure class.

Proposition 5.1 (Source structure and low costs)**.** *Every $F \in \mathcal{M}_d$, for $d \in \{4, 6, 8, 10, 12\}$, is connected, d-regular, factor-critical, and well-covered. Its order is $2k + 1 = 2r + s$ and its gap is one. Its low costs are*

$$c_0(F) = k + E, \quad c_1(F) = k + E - \epsilon = 2k + 1 - r + \beta, \quad c_2(F) = k + E - \delta = 2k + 1 - r + p.$$

The excess floors in degree order $4, 6, 8, 10, 12$ are $14, 6, 10, 12, 10$.

At degree six,

$$E \geq 6, \quad 2 \leq \lambda \leq E - 4, \quad p \geq 2, \quad 2 \leq \epsilon < \delta \leq \lambda + 1, \quad 1 \leq s \leq \epsilon,$$

and s is odd. At degree eight,

$$E \geq 10, \quad 2 \leq \epsilon \leq \delta \leq \lambda \leq \lfloor (E - 4)/2 \rfloor, \quad \lambda \geq 3, \quad p \geq \lambda + 2, \quad q_8 \geq 1, \quad p \geq q_8 + \delta + 1.$$

At degree ten, $E \geq 12$, $\lambda \geq 3$, and $p \geq 5$ in both sectors. At degree twelve, $E \geq 10$, $\lambda \geq 3$, $p \geq 5$, and $\delta > 0$. For degrees ten and twelve, $0 \leq \epsilon \leq \delta \leq \lambda + 1$ and s is positive odd.

A complete proof is recorded in [Rec]; the supporting development is collected in [II]. The displayed low-cost identities themselves follow from the definitions and $2k + 1 = 2r + s$: $2k + 1 - r + p = k + E - \delta$ and $\beta - p = \delta - \epsilon$.

Proposition 5.2 (Induced-minimal restrictions in degrees six and eight)**.** *Let H be an induced-minimal T_- failure of maximum degree $d \in \{6, 8\}$. Then H is a totally critical closed d-lump of gap one. Write $|V(H)| = 2k + 1$, $E = \mathrm{ex}(H)$, and $v = \max(0, \lfloor (2E - 3)/(d - 2) \rfloor)$. Then*

$$E \geq 4, \qquad \alpha(H) \geq \left\lceil \frac{3E + 5}{4} \right\rceil \geq 5, \qquad k \geq h_-(d) + 2E - 1 + 2v.$$

For every vertex x and $0 \leq i \leq d$,

$$c_i(H - x) = c_i(H) - 1, \qquad \mathrm{ex}(H - x) = E - 1, \qquad \mathrm{gap}(H - x) = 0.$$

For every maximal independent set S, put $a_S = |S|$, $G = H - S$, and $\eta_S = \nu(G) - (k + 1 - a_S)$. Then

$$a_S \geq 2, \quad \Delta(G) \leq d - 1, \quad 0 \leq \eta_S \leq \min(E - 1, \lfloor (a_S - 1)/2 \rfloor),$$
$$\mathrm{ps}(G) = E - 1 - \eta_S, \quad c_i(H) = a_S + c_i(G) \ (1 \leq i < d), \quad c_d(H) = 1,$$
$$c_0(H) = a_S + \nu(G) + \mathrm{ps}(G), \qquad \mathrm{gap}(G) \in \{0, 1\}.$$

A complete proof is recorded in [Rec]; the supporting development is collected in [II].

Proposition 5.3 (Sextic native rows and processes)**.** *Let $F \in \mathcal{M}_6$. Native rows exist. Every native row is well-covered with independence number a and has form*

$$A = zK_1 + bP_2 + tP_4 + n_3C_3 + n_4C_4 + n_5C_5 + n_7C_7,$$

where the component counts satisfy

$$n_3 + n_5 + n_7 = p, \quad b + t = \epsilon, \quad b + 2t + 2n_4 + n_5 + 2n_7 = \delta, \quad z = a - p - \delta.$$

Thus $q_{\mathrm{def}} = t + 2n_4 + n_5 + 2n_7$ and $e(A) = 3p + 2\delta - \epsilon$. Every maximal independent set of A is maximum in F. The exact source ladder is

$$(c_0, c_1, c_2, c_3, c_4, c_5)(F) = (k + E, k + E - \epsilon, k + E - \delta, r, a + p + \epsilon, a).$$

For $J = F - V(A)$,

$$\Delta(J) \le 3, \quad \mathrm{ps}(J) = 0, \quad \alpha(J) = a, \quad \nu(J) = p + s + \delta,$$
$$c_1(J) = p + s + \delta - \epsilon, \quad c_2(J) = p + s, \quad c_i(J) = 0 \ (i \ge 3).$$

Every source prefix and terminal has independence number a. Every complete source process has word

$$5^{p+\epsilon} 4^{p+q_{\mathrm{def}}} 3^s.$$

Its initial independent set and full first two phases induce a native row. Every native row admits such an early realization, after which any prescribed continuation on its unchanged complement may be appended.

Put $U = \{v \in V(J) : d_J(v) = 3\}$ and $\Gamma = J[U]$. The complete cubic phases are exactly the maximal independent sets of Γ, each of size s, in any order. For any such phase X, the remainder $R = J - X$ is native; A, X, R partition $V(F)$, X is independent, and every $x \in X$ has three neighbors in each row. Deleting the same X in any order from $F - V(R)$ returns to A.

Every cubic has exactly one odd whole A-component neighbor, and every owned odd component has exactly one cubic owner. Let $Y \subseteq V(J) \setminus U$, and let $m(Y)$ count the owned odd components touched by Y. For every missing pair $b_R + b_B = c_0(F) - 1$, put $j = u - b_R$ and $v_6 = \lfloor (2E - 3)/4 \rfloor$. Then

$$j \ge p + s + \delta + 1 + v_6 = E + \lambda + 2 - \delta + v_6 \ge E + 1 + v_6 \ge 7.$$

If $|Y| + m(Y) \le j$, one matching assigns all of Y to distinct free odd components through exposed ports, simultaneously assigns the owners of the touched owned components, and is maximum internally on every whole row component. No independence of Y is required.

A complete proof is recorded in [Rec]; the supporting development is collected in [II].

The explicit replay and cubic-phase steps. For a fixed native row, number each path consecutively and each cycle cyclically. Choose the initial independent set and degree-five set componentwise as follows:

component	S_A	T_5
K_1	$\{1\}$	$\varnothing$
P_2	$\{1\}$	$\{2\}$
P_4	$\{1,3\}$	$\{4\}$
C_3	$\{1\}$	$\{2\}$
C_4	$\{1,3\}$	$\varnothing$
C_5	$\{1,3\}$	$\{4\}$
C_7	$\{1,3,5\}$	$\{6\}$

Put $T_4 = V(A) \setminus (S_A \cup T_5)$. The component identities give $|S_A| = a$, $|T_5| = p + \epsilon$, and $|T_4| = p + q_{\mathrm{def}}$. The prescribed-row realization input verifies that these are the complete phases in every internal order.

In a subcubic graph, roots deleted at current degree three are original cubics and are pairwise nonadjacent. Completeness means that every other original cubic has a neighbor among them. Thus they form a maximal independent set of Γ. Conversely, deleting any maximal independent set of Γ in any order is legal and leaves maximum degree at most two. The process profile gives its size s. The stated three-neighbor identities then make deletion of the same roots from the opposite complement a legal reverse continuation. □

Proposition 5.4 (Native rows in degrees ten and twelve)**.** *Let $F \in \mathcal{M}_d$ with $d = 10$ or 12. Native rows exist and every terminal is native. Every source prefix has independence number a. Every native A has independence number a, $e(A) = 3p + 2\delta - \epsilon$, exactly p odd-cycle components, and exactly ϵ positive even-order path components.*

For $d = 10$, the permitted components are $K_1, P_2, P_3, P_4, C_3, C_4, C_5, C_7$. For $d = 12$, they are $K_1, P_2, P_4, C_3, C_4, C_5, C_7$, and every native row is well-covered. The early two phases have sizes $p + \epsilon, p + q_{\mathrm{def}}$. Together with the initial independent set they induce a native row. Every native row has such an early realization.

For $J = F - V(A)$, put $m = p + \delta$. Then

$$|V(J)| = u, \quad \alpha(J) = a, \quad \nu(J) = m + s, \quad \mathrm{ps}(J) = 0,$$

and J is $(d-3)$-good. For $1 \le i \le d-3$, $c_i(J) = c_i(F) - r$. Also

$$c_1(J) = m + s - \epsilon, \qquad c_2(J) = m + s - \delta,$$

$$e(J) = 3p + 2\delta - \epsilon + \frac{d}{2}s, \qquad \sum_{i \ge 3} c_i(J) = \left(\frac{d}{2} - 3\right) s.$$

Every complete continuation on this same J is an actual source continuation. It has s roots, deletion-degree sum $ds/2$, and terminal native row R. If X is its whole root set, then A, X, R partition the source,

$$\Delta(F[X]) \le d - 6, \qquad e_F(X, V(A)) = e_F(X, V(R)) = \frac{d}{2}s - e(F[X]).$$

The top source costs are

d	$c_{d-3}(F)$	$c_{d-2}(F)$	$c_{d-1}(F)$
10	r	$a + p + \epsilon$	a
12	r	$a + p + \epsilon$	a

and $a + p + \epsilon = r - \beta$.

A complete proof is recorded in [Rec]; the supporting development is collected in [II].

Proposition 5.5 (Paired phases and exact costs)**.** *For degree ten, every native-row complement has a complete phase profile*

$$(|B_7|, |B_6|, |B_5|, |B_4|, |B_3|) = (x, y, s - 2x - 2y, y, x),$$

where the middle length is positive odd. Its source costs at indices three through seven are

$$(r + s - x,\ r + s - x - y,\ r + x + y,\ r + x,\ r).$$

For degree twelve, there are counts t_9, t_8, t_7 depending only on F such that every native-row complement has word

$$9^{t_9} 8^{t_8} 7^{t_7} 6^{s - 2(t_9 + t_8 + t_7)} 5^{t_7} 4^{t_8} 3^{t_9},$$

with positive odd middle length. Its costs at indices three through nine are

$$(r + s - t_9,\ r + s - t_9 - t_8,\ r + s - t_9 - t_8 - t_7,$$
$$r + t_9 + t_8 + t_7,\ r + t_9 + t_8,\ r + t_9,\ r).$$

In particular

$$c_3 + c_8 = c_4 + c_7 = c_5 + c_6 = 2r + s.$$

In degree twelve, if the actual roots are $x_1, \dots, x_s$ with deletion degrees $d_1, \dots, d_s$, then deletion of $x_s, \dots, x_1$ from the opposite complement is legal, returns exactly to A, and deletes x_i at degree $12 - d_i$. Moreover $d_i + d_{s+1-i} = 12$. Every phase is independent, and its internal order is arbitrary. For $X_l = \{x_1, \dots, x_l\}$ and $G_l = F[V(A) \cup X_l]$,

$$\mathrm{ps}(G_l) = 0, \quad \nu(G_l) = \nu(A) + l, \quad c_j(G_l) = c_j(A) + |\{i \le l : j < 12 - d_i\}| \quad (j \ge 1).$$

A complete proof is recorded in [Rec]; the supporting development is collected in [II].

6. The uniform exclusion mechanism

The goal of this section is to turn the source profiles into complete certificates whose totals are one below an exact source cost.

6.1. **Restoration and forced matching ranks.**

Lemma 6.1 (Restoring omitted vertices)**.** *Let $H = F - S$ be induced, $N = |S|$, and let $M_0, \dots, M_{D-1}$ be pairwise edge-disjoint matchings in H, where $D \geq 1$. Put*

$$K = \bigcup_{i<D} M_i, \qquad P = E(H) \setminus K, \qquad m = \nu(P).$$

If $N \geq \max(0, m-1)$, then every capacity split $A + B = D$ and every budget split $x + y = N + m$ has a certificate on F with $Q_R \cup Q_B = K$.

If the paid graph is exactly one cycle of order $2e-1$, g disjoint edges, and isolates, it suffices that $N + g \geq e - 2$. The resulting total is $N + g + e - 1$.

Proof. Put the first A matching sets into Q_R and the remaining B into Q_B. A vertex meets at most one edge in each matching, so the capacities hold.

Order S and assign each original edge meeting S to its earliest endpoint in that order. This partitions every edge outside H into N indexed stars, possibly empty. An edge with two omitted endpoints is assigned once.

If $m = 0$, distribute these stars according to x, y. If $m \geq 1$, then $N + m \geq 2m - 1$, so one integer budget is at least m. If $x \geq m$, put all of P and $x - m$ stars in P_R, and put the remaining y stars in P_B. The number $x - m$ lies between zero and N. Restricting a matching to the edge pieces proves both matching bounds. If $y \geq m$, interchange the paid parts.

For the cycle assertion, combine the g disjoint paid edges with the N boundary stars. These are $N + g$ indexed pieces of matching number at most one. The cycle has matching number $e - 1$. The inequality $N + g \geq e - 2$ gives total at least $2(e-1) - 1$, so the same assignment applies. Shared vertices between boundary stars and other pieces do not affect the restriction bound. □

Proposition 6.2 (Six-prefix restoration and its sharp threshold)**.** *Fix six pairwise edge-disjoint matchings in an induced $H = F - S$. Let P_i be the remainder after the first i matchings. If $N = |S| \geq \max(0, \nu(P_1) - 1)$, then $c_i(F) \leq N + \nu(P_i)$ for every $1 \leq i \leq 6$, with exactly those i matchings unpaid. It is enough that the second matching is maximum in P_1 and $N \geq \max(0, |M_1| - 1)$.*

If P_6 is one cycle of order $2e - 1$, gP_2, and isolates, and $N + g \geq e - 2$, then $c_6(F) \leq N + g + e - 1$. If also

$$|V(F)| = 2r + s, \quad |V(H)| = r + h, \quad h = s + \mu + \kappa + e, \quad \mu - g = L, \quad \kappa + L = \beta,$$

then the total equals $r - \beta - 1$.

The threshold $N + g \geq e - 2$ is sharp for a universal guarantee with the exact unpaid union fixed. This is not a lower bound on unrestricted c_6.

Proof. Edge deletion gives $\nu(P_i) \leq \nu(P_1)$, so Lemma 6.1 applies at every prefix. For the cycle total, substitute the actual omitted count:

$$N + g + e - 1 = r + s - h + g + e - 1 = r - (\mu - g) - \kappa - 1 = r - \beta - 1.$$

For sharpness, let $t = e - 1$ and $L_0 = N + g < t - 1$. Take a disjoint union of $K_{6,6}$, a cycle of order $2t + 1$, g single edges, and N further edges $s_j z_j$. Omit the vertices s_j. Decompose $K_{6,6}$ into the six nonempty matchings $\{a_i b_{i+j \bmod 6} : 0 \leq i < 6\}$. Their exact complement in H has the required form.

At budgets $x = t - 1$, $y = L_0 + 1$, both budgets are below t. Thus the cycle must have edges in both paid parts. Each proper cycle-edge subgraph is a union of paths, whose matching number is at least half its edge count. The two cycle restrictions therefore have matching numbers summing to at least $t + 1$. The L_0 other disjoint paid edges add exactly L_0. This exceeds the total budget $L_0 + t$. The contradiction is independent of the capacity split. □

Lemma 6.3 (Three successive row matchings)**.** *Let $A \subseteq H = F - S$ be induced, put $W = V(H) \setminus V(A)$, $h = |W|$, and $N = |S|$. Assume A consists of isolates, paths, and cycles from $K_1, P_2, P_3, P_4, C_3, C_4, C_5, C_7$. Suppose its matching number is m, it has p odd cycles and ϵ positive even-order paths, and set $\beta = m - \epsilon$. Assume*

$$\nu(H) = m + h, \quad N \geq \max(0, \beta + h - 1), \quad c_1(F) = N + h + \beta, \quad c_2(F) = N + h + p.$$

Choose any successive maximum matchings M_0, M_1, M_2. Then their ranks are $m+h, \beta+h, p+h$. Each matches every W-vertex into A and uses no W-internal edge. Their union contains every A-edge. After M_0, the row is $pP_3 + (\beta - p)P_2$ and isolates. After M_1, it is pP_2 and isolates. The restored totals at indices one and two equal the source costs.

Proof. Any matching in H has at most m row edges and at most h other edges. Equality forces both bounds to be attained. Each nonrow edge then uses exactly one W-vertex. The row restriction is maximum on each component, and crossing endpoints are exposed ports in distinct odd components.

Removing a maximum matching from an odd cycle leaves one P_3 and disjoint edges. Removing one from P_3 leaves one edge; removing one from P_4 leaves its middle edge; removing one from C_4 leaves two edges. Component counting gives the asserted first residual row of matching number β. Hence $\nu(P_1) \le \beta + h$. Lemma 6.1 and the first source equality force equality. Every maximum M_1 therefore saturates W into A and is maximum internally. The second row residual consists of p edges. The same argument at $c_2(F)$ gives rank $p+h$ and forces M_2 to contain all those row edges. All three matchings preserve $H[W]$. □

6.2. **Degree twelve.**

Proposition 6.4 (Shortest-cycle reserve)**.** *Let $F \in \mathcal{M}_{12}$. Choose any native A, any complete continuation on its complement, and its terminal R and full root set X_{root}. Put $D_2 = \{v \in V(R) : d_R(v) = 2\}$. Choose any maximum independent set T_0 of $R[D_2]$ and a shortest whole odd cycle C of R. Set*

$$e_C = (|V(C)|+1)/2, \quad Y = T_0 \cup V(C), \quad W = X_{\mathrm{root}} \cup Y, \quad H = F[V(A) \cup W], \quad h = |W|, \quad N = |V(F) \setminus V(H)|.$$

These choices exist, and, with $v_{12} = \lfloor (2E-3)/10 \rfloor$,

$$|T_0| = \beta, \quad 2 \le e_C \le 4, \quad e_C \le \epsilon + v_{12} + 1, \quad h = s + \beta + e_C, \quad N = r - \beta - e_C.$$

The graph $R[Y]$ is exactly C and isolates. Moreover

$$N - (\beta + h - 1) \ge 3\epsilon + 2v_{12} + 2 - 2e_C = \epsilon + 2(\epsilon + v_{12} + 1 - e_C) \ge 0,$$

and $N \ge e_C - 2$. For every missing pair (x, y),

$$u - x - h, \ u - y - h \ge \epsilon + v_{12} + 1 - e_C \ge 0.$$

Proof. The native component list has $p \ge 5$ odd cycles, of orders three, five, or seven. A maximum independent set on their degree-two vertices, on each C_4, and on each internal P_4 edge has size β. Completing C adds e_C vertices. Every other selected vertex remains isolated in $R[Y]$. This proves the cardinalities and $2 \le e_C \le 4$.

The missing-allocation order estimates needed here are

$$u - x - h, \ u - y - h \ge \epsilon + v_{12} + 1 - e_C, \qquad N - (\beta + h - 1) \ge 3\epsilon + 2v_{12} + 2 - 2e_C.$$

They leave only the reserve inequality to check.

Suppose $e_C > \epsilon + v_{12} + 1$. Since $v_{12} \ge 1$, the case $e_C = 2$ is impossible. If $e_C = 3$, then $\epsilon = 0$, $v_{12} = 1$, and $E \in \{10, 11\}$. No terminal triangle is present, so $p + \delta = \nu(R) \ge 2p + \epsilon$. Together with $\delta \le \lambda + 1 = E - 1 - p$, this forces

$$(E, p, \epsilon, \delta, \lambda) = (11, 5, 0, 5, 4).$$

The required boundary theorem says that every native row at these parameters contains a triangle, a contradiction.

If $e_C = 4$, then all odd cycles are heptagons and $\delta \ge 2p + \epsilon$. Thus $E \ge 3p + \epsilon + 1 \ge 16$. The only possible reserve failure is $(E, p, \epsilon, \delta, \lambda, v_{12}) = (16, 5, 0, 10, 9, 2)$. Here $r = k$, $s = 1$. The required order estimates give simultaneously

$$23k < 100E - 100 - 50\lambda = 1050, \qquad k = r \ge s + 3p + 3\delta + 1 + 2v_{12} = 51,$$

which is impossible. The reserve follows. Finally $\beta \ge p \ge 5$ and $s \ge 1$ give $\beta + h - 1 = s + 2\beta + e_C - 1 \ge e_C$, so the omission bound implies $N \ge e_C - 2$. □

A complete proof is recorded in [Rec]; the supporting development is collected in [II].

Proposition 6.5 (The degree-twelve matching prefix)**.** *Fix any $F \in \mathcal{M}_{12}$, native A, and missing pair (x, y). Put $j_* = u - \max(x, y)$ and $m = p + \delta$. Then*

$$1 \le s + p + \delta + v_{12} + 1 \le j_* \le u.$$

For every $W \subseteq V(F) \setminus V(A)$ with $h = |W| \le j_*$, the induced $H = F[V(A) \cup W]$ has $\nu(H) = m + h$. For $N = u - h$,

$$N - (\beta + h - 1) = |x - y| + 2(j_* - h) + \epsilon \ge 0.$$

Every three successive maximum matchings on H have the conclusions of Lemma 6.3. Every retained vertex of complement degree nine uses its unique odd-owner edge in M_0 and its other two A-edges in M_1, M_2. All these fixed edges remain in the unpaid prefix. For every independent subset X of those retained degree-nine vertices, $V(A) \cup X$ is independent in the third residual.

A complete proof is recorded in [Rec]; the supporting development is collected in [II].

The remaining deductions. The identity follows from $x + y = u + m - 1$ and $\beta = m - \epsilon$. The source costs are $c_1(F) = u + \beta$ and $c_2(F) = u + p$, so Lemma 6.3 applies. A complement-degree-nine vertex has three A-neighbors by twelve-regularity. The three crossing matchings use three distinct such edges. Its only possible exposed odd-component contact is its owner edge, so that edge is forced in M_0. The row edges and every retained degree-nine boundary are exhausted. Independence of X gives the last assertion. □

Proposition 6.6 (The ten-matching lift)**.** *Use the graph H and selector of Proposition 6.4. Suppose three edge-disjoint matchings on H each saturate W into A and together contain every A-edge. Assume $N \ge \max(0, h - t_9 - 2)$. Successive maximum matchings at indices three through nine have ranks*

$$\begin{aligned}&(h - t_9,\ h - t_9 - t_8,\ h - t_9 - t_8 - t_7,\\ &\ h - s + t_9 + t_8 + t_7,\ h - s + t_9 + t_8,\ h - s + t_9,\ h - s).\end{aligned}$$

Their restored ranks equal $c_i(F)$ at those indices. The final maximum matching is unique. After all ten matchings, the entire remaining edge set is $E(C)$. When the stronger omission bound $N \ge \max(0, \beta + h - 1)$ holds, every prefix from index three through ten has a full-source certificate at total $N + \nu(P_i)$. In particular

$$c_{10}(F) \le N + e_C - 1 = r - \beta - 1 = c_{10}(F) - 1.$$

Proof. Put

$$I_i = V(A) \cup \bigcup_{12-i \le j \le 9} B_j, \qquad C_i = V(H) \setminus I_i \qquad (3 \le i \le 9).$$

The first three matchings clear A and all three A-contacts of B_9. Thus I_3 is independent. The phase counts in Proposition 5.5 give the displayed cover sizes $b_i = |C_i|$, and $N + b_i = c_i(F)$.

Suppose I_i is independent. Then $\nu(P_i) \le b_i$. If $\nu(P_i) \le b_i - 1$, the weak omission bound implies $N \ge \nu(P_i) - 1$, since $b_i \le b_3$. Lemma 6.1 would give $c_i(F) \le N + \nu(P_i) < N + b_i = c_i(F)$. Hence $\nu(P_i) = b_i$. Every maximum matching saturates the cover and crosses into I_i.

Every vertex of B_{11-i} has $i+1$ original neighbors in I_i. Exactly i have been used, one by each preceding matching. Its unique remaining earlier edge is therefore forced in M_i. The next side becomes independent. This proves the induction through the physical root layers. At index nine the side is $X = V(A) \cup X_{\mathrm{root}}$ and the cover is Y.

Every $y \in Y$ has ten original neighbors in X. The first nine matchings remove nine distinct such edges. Its last edge is $y\phi(y)$. A maximum matching of rank $|Y|$ must contain all these edges. Its existence proves injectivity of ϕ, and the matching is unique. Deleting it leaves precisely $R[Y]$, since no preceding matching used a Y-internal edge.

Under the stronger omission bound, every paid rank from index three on is at most h and $N \ge h - 1$. Lemma 6.1 supplies all the stated certificates. For the last one, its cycle form also applies directly because $N \ge e_C - 2$. The exact omitted count and $c_{10}(F) = r - \beta$ give the contradiction. □

Theorem 6.7 (Degree twelve)**.** *The class $\mathcal{M}_{12}$ is empty. Equivalently, every graph of maximum degree exactly twelve whose proper subgraphs satisfy T_- also satisfies T_-.*

Proof. Suppose $F \in \mathcal{M}_{12}$. Proposition 6.4 supplies one actual row, continuation, selector, and missing pair with $h \le u - \max(x, y)$. Proposition 6.5 supplies the three compatible matchings on that same induced H. Moreover

$$(\beta + h - 1) - (h - t_9 - 2) = \beta + t_9 + 1 \ge 1.$$

Thus the weak lift margin holds. Proposition 6.6 gives a full certificate of total $c_{10}(F)-1$, a contradiction. No row, selector, phase, or omission premise remains beyond membership in $\mathcal{M}_{12}$. □

6.3. **Degree eight.**

Proposition 6.8 (The octic cubic-entry selector)**.** *Let $F \in \mathcal{M}_8$, fix any native A and any continuation on its complement, and put $q = q_8$. Its word is $5^q 4^{s-2q} 3^q$, with positive odd quartic length. Let B_5, B_4 be its full first two phases and $K = F - (V(A) \cup B_5 \cup B_4)$. Then $|B_5| = q$, $|B_4| = s - 2q$, $|V(K)| = r + q$, and $\alpha(K) = a$.*

Put $U = \{v : d_K(v) = 3\}$, $\Gamma = K[U]$, $U_0 = \{v : d_\Gamma(v) = 0\}$, and $\mu = \nu(\Gamma)$. The whole components of K belong to

$$K_1, P_2, P_4, C_3, C_4, C_5, C_7, B_{33}, B_{35}, B_{55}, S_6, \mathrm{cor}(P_3), \mathrm{cor}(P_4), \mathrm{cor}(P_6), \mathrm{cor}(C_4).$$

Here B_{ij} joins two disjoint cycles by one bridge, $\mathrm{cor}(B)$ attaches a private leaf at every vertex of B, and S_6 is a square with private leaves at two adjacent vertices.

Let $n(T)$ count whole components of type T, and set

$$g = n(\mathrm{cor}(P_6)) + 2n(\mathrm{cor}(C_4)),$$
$$L = n(B_{33}) + n(B_{35}) + n(B_{55}) + n(S_6) + n(\mathrm{cor}(P_4)) + n(\mathrm{cor}(P_6)).$$

Then Γ is a union of K_1, K_2, P_4, C_4 and

$$|U| = q + \mu, \quad \mu - g = L, \quad 0 \le g \le q \le \epsilon.$$

Every maximum matching of Γ covers all nonisolated vertices and leaves exactly g disjoint edges.

Put $D = \{v : d_K(v) = 2,\ N_K(v) \cap U = \varnothing\}$. The graph $K[D]$ consists of all whole ordinary cycles, the middle edge of each ordinary P_4, and the opposite internal edge of each pentagon shore of a barbell. For any maximum independent T_0 of $K[D]$, write $\kappa = |T_0|$. Then

$$\kappa = n(P_4) + n(C_3) + 2n(C_4) + 2n(C_5) + 3n(C_7) + n(B_{35}) + 2n(B_{55}), \qquad \kappa + L = \beta.$$

There are at least $p - \delta \ge 2$ whole ordinary triangles in K. Choose any such C and put $Y = T_0 \cup V(C)$. Then $|Y| = \kappa + 2$, every Y-vertex has degree two and avoids U, and $K[Y]$ is C and isolates. For every maximum Γ-matching M_Γ,

$$K[U \cup Y] - M_\Gamma = C \sqcup gP_2 \sqcup \textit{isolates}.$$

For $W = B_5 \cup B_4 \cup U \cup Y$ and $H = F[V(A) \cup W]$,

$$h = s + \beta + g + 2, \quad N = u - h, \quad N + g + 1 = r - \beta - 1 = c_6(F) - 1.$$

A complete proof is recorded in [Rec]; the supporting development is collected in [II]. The classification excludes whole K_4 components. The source costs are $c_3 = r + s - q$, $c_4 = r + q$, $c_5 = r$, $c_6 = r - \beta$. The classification, prescribed-row realization, and source costs hold for every prescribed continuation.

The selector calculation from the component input. The cubic pieces are K_2 for barbells, S_6, and $\mathrm{cor}(P_4)$; P_4 for $\mathrm{cor}(P_6)$; C_4 for $\mathrm{cor}(C_4)$; and K_1 for $\mathrm{cor}(P_3)$. Removing a maximum matching leaves respectively zero, one, two, and zero edges. This proves $\mu - g = L$ and the residual assertion for Γ.

Inspecting degree-two vertices gives the stated graph $K[D]$. On a pentagon shore $v_0v_1v_2v_3v_4v_0$ with bridge endpoint v_0, only v_2, v_3 avoid the cubic set. Maximum independent sets restrict maximally to each component. The independence numbers of C_3, C_4, C_5, C_7 are $1, 2, 2, 3$, which gives κ.

For completeness, the local terminal counts after any full cubic phase are listed below. The columns are the number o_T of odd cycles, terminal matching number m_T, number f_T of positive even paths, and

the contributions to κ, L.

T	o_T	m_T	f_T	κ_T	L_T
K_1	0	0	0	0	0
P_2	0	1	1	0	0
P_4	0	2	1	1	0
C_3	1	1	0	1	0
C_4	0	2	0	2	0
C_5	1	2	0	2	0
C_7	1	3	0	3	0
B_{33}	1	2	1	0	1
B_{35}	1	3	1	1	1
B_{55}	1	4	1	2	1
S_6	0	2	1	0	1
$\mathrm{cor}(P_3)$	0	2	2	0	0
$\mathrm{cor}(P_4)$	0	3	2	0	1
$\mathrm{cor}(P_6)$	0	4	3	0	1
$\mathrm{cor}(C_4)$	0	2	2	0	0

A barbell loses one bridge endpoint and leaves the opposite cycle and an even path. The path-corona phases delete independent internal bases; surviving bases and their leaves form P_2 or P_4 components. The square corona leaves two edges. These descriptions verify the terminal columns. Each row satisfies $m_T - f_T = \kappa_T + L_T$. Summing gives $\kappa + L = p + \delta - \epsilon = \beta$.

For every type except ordinary C_3, the difference $m_T - o_T$ is at least o_T. Thus $\delta \geq p - n(C_3)$. Since $p \geq \lambda + 2$ and $\delta \leq \lambda$, at least two ordinary triangles occur. Completing one adds two vertices to T_0 and creates only its triangle edges. There are no U–Y edges. The residual and the final cardinality calculation follow:

$$h = q + (s - 2q) + (q + \mu) + (\kappa + 2) = s + \beta + g + 2.$$

□

Proposition 6.9 (The octic matching prefix)**.** *Let $F \in \mathcal{M}_8$, let A be any native row, and choose integers $0 \leq g \leq \min(q_8, \epsilon)$ and $1 \leq e_{\mathrm{sel}} \leq 4$. For every exterior set of size $h = s + \beta + g + e_{\mathrm{sel}}$, the induced H satisfies*

$$\nu(H) = p + \delta + h, \qquad N = u - h \geq \max(0, \beta + h - 1).$$

An initial maximum matching may be chosen after prescribing every retained complement-degree-five owner edge, compatible internal matchings exposing those ports, perfect matchings on even row components, and maximum matchings on odd components with no retained neighbor. Omitted owners need not be reserved. Every such initial choice followed by arbitrary successive maximum matchings has the conclusions of Lemma 6.3, simultaneously for every native row contained in H. Every retained complement-degree-five boundary is exhausted. All prescribed initial edges remain unpaid.

A complete proof is recorded in [Rec]; the supporting development is collected in [II].

Proposition 6.10 (The six-matching lift)**.** *Use the graph H of Proposition 6.8. Suppose three edge-disjoint matchings each saturate W into A and clear every A-edge, and assume $N \geq \max(0, h - q_8 - 1)$. Then arbitrary successive maximum matchings at indices three and four have ranks $h - q_8$ and $h - s + q_8$. At index five the rank is $h - s$. Every maximum fifth matching consists of a maximum Γ-matching and edges saturating Y into $V(A) \cup B_5 \cup B_4$. Its Γ part may be replaced by any maximum Γ-matching. The entire sixth remainder is*

$$E(P_6) = E(K[U \cup Y]) \setminus M_\Gamma,$$

with A, B_5, B_4 isolated. At indices three through six the full restored certificates have totals $N + \nu(P_i)$.

Proof. After the prefix, $A \cup B_5$ is independent: each B_5 vertex has exactly three A-neighbors and all were used. Its complementary cover has size $h - q_8$. Restoration and $c_3(F) = r + s - q_8$ force equality in this cover bound. Every maximum M_3 saturates the cover across it. Each B_4 vertex had four earlier neighbors and has one left, so this edge is removed.

Now $A \cup B_5 \cup B_4$ is independent and $U \cup Y$ is a cover of size $h - s + q_8$. The equality $c_4(F) = r + q_8$ forces the next rank and crossing saturation. A cubic U-vertex had five earlier neighbors; its last one is removed. A Y-vertex had six and has one left. No $K[U \cup Y]$ edge has been removed.

Every remaining edge outside Γ meets Y, so the next matching rank is at most $\mu + |Y| = h - s$. Restoration and $c_5(F) = r$ force equality. The Γ restriction is maximum and covers all of $U \setminus U_0$. Since Y has no neighbor in U_0, the other matching edges go to the earlier side and exhaust its last Y-contacts. This proves the literal remainder. All paid ranks decrease from the first cover rank, so the stated omission bound justifies restoration at every used index. □

Theorem 6.11 (Degree eight)**.** *The class $\mathcal{M}_8$ is empty.*

Proof. For an arbitrary alleged source, Proposition 6.8 supplies an actual selector with $h = s + \beta + g + 2$ and $0 \le g \le \min(q_8, \epsilon)$. Apply Proposition 6.9 with $e_{\mathrm{sel}} = 2$. Its stronger omission bound implies the hypothesis of Proposition 6.10. The latter gives six matchings on this same H and a remainder consisting of the selected triangle, g disjoint edges, and isolates. The selector assertion holds for every maximum Γ-matching, so it applies to the one delivered by the lift. Proposition 6.2, with $e = 2$, gives

$$c_6(F) \le N + g + 1 = r - \beta - 1 = c_6(F) - 1.$$

The certificate includes every omitted-incident edge. This contradiction excludes the entire class. □

6.4. **Degree six.**

Proposition 6.12 (The sextic physical selector)**.** *Let $F \in \mathcal{M}_6$, fix any native A, and put $J = F - V(A)$, $U = \{v : d_J(v) = 3\}$, $\Gamma = J[U]$, and $U_0 = \{v : d_\Gamma(v) = 0\}$. The whole component list is the same fifteen-type list used in Proposition 6.8. Set $b = |U_0|$, $\mu = \nu(\Gamma)$, and define g, L by that proposition's component formulas. Then*

$$|U| = 2s - b = s + \mu, \qquad \mu = s - b, \qquad \mu - g = L.$$

Let D consist of the degree-two vertices avoiding U. Its induced graph has the same component description as in Proposition 6.8. For any maximum independent T_0 of $J[D]$, put $\kappa = |T_0|$. Then its displayed component formula still holds and

$$\kappa + L = p + q_{\mathrm{def}} = \beta, \qquad p = n(C_3) + n(C_5) + n(C_7) + n(B_{33}) + n(B_{35}) + n(B_{55}) \ge 2.$$

Choose any whole ordinary odd cycle or any one shore of a whole barbell. Call it C, put

$$Z = V(C) \setminus (U \cup T_0), \quad e_C = |Z|, \quad Y = T_0 \cup Z.$$

Then

$$e_C = \lfloor |V(C)|/2 \rfloor + 1 \le 4, \qquad |Y| = \kappa + e_C \le \beta + 4.$$

Every Y-vertex has J-degree two and avoids U_0. For every missing pair, with $j = u - b_R$,

$$|Y| + m(Y) \le |Y| + |U| \le j.$$

For every maximum matching M_Γ,

$$J[U \cup Y] - M_\Gamma = C \sqcup gP_2 \sqcup \textit{isolates}, \qquad \nu = g + e_C - 1.$$

A complete proof is recorded in [Rec]; the supporting development is collected in [II].

The physical selector and booking calculation. The inspection of degree-two vertices and the local cubic matching calculation are those in the proof of Proposition 6.8, applied to the stated sextic component input. Their contributions to $\kappa + L$ are

$$0, 0, 1, 1, 2, 2, 3, 1, 2, 3, 0, 1, 1, 0, 1$$

in the order $K_1, P_2, P_4, C_3, C_4, C_5, C_7, B_{33}, B_{35}, B_{55}, \mathrm{cor}(P_3), \mathrm{cor}(P_4), \mathrm{cor}(P_6), \mathrm{cor}(C_4), S_6$. The sextic component identities give their sum β.

On an ordinary odd cycle of order $2v + 1$, T_0 uses v vertices. On a triangle shore it uses none, and on a pentagon shore it uses one vertex of the internal edge. Each shore has one cubic bridge endpoint. These counts prove the formula for e_C. The only possible cubic neighbor of an added shore vertex is its bridge endpoint, which is nonisolated in Γ. Thus Y avoids U_0.

Put $v_6 = \lfloor (2E-3)/4 \rfloor$. The booking bound in Proposition 5.3 gives

$$j - |U| - \kappa \geq \epsilon - s + 1 + v_6 + b + L.$$

Here $v_6 \geq 2$ and $\epsilon \geq s$, so the right side is at least three. This covers $e_C \leq 3$. If $e_C = 4$ and $\epsilon > s$, it is at least four. If $\epsilon = s$, then s is odd and at least three. Writing $t = (s-1)/2$ gives $\lambda = q_{\mathrm{def}} + \epsilon + t - 1$ and $E = p + q_{\mathrm{def}} + \epsilon + t + 1 \geq 8$, hence $v_6 \geq 3$. The bound is again four. Since every owned component has one cubic owner, $m(Y) \leq |U|$, proving the booking assertion.

Every maximum Γ-matching deletes a barbell bridge, which is a whole K_2 component of Γ. Thus a selected shore has no surviving attachment to the other shore. Every selected vertex outside C is independent and avoids U. The remaining nonisolated edges outside C are exactly the g residual cubic edges. This proves the entire remainder assertion. □

Proposition 6.13 (Four compatible sextic matchings)**.** *Use the notation of Proposition 6.12. More generally, let $Y \subseteq V(J) \setminus U$ satisfy $|Y| \leq \beta + 4$ and $|Y| + m(Y) \leq j$ for one missing pair. Put $h = |U| + |Y|$, $H = F[V(A) \cup U \cup Y]$, and $N = u - h$. There is one matching M_0 containing any prescribed cubic owner edges, assigning Y to distinct free odd components, and maximum internally on every row component. For any such choice, three successive maximum matchings have paid ranks*

$$\nu(P_1) = \beta + h, \qquad \nu(P_2) = p + h, \qquad \nu(P_3) = \mu + |Y| = h - s.$$

The first three matchings clear every row edge and every U–A edge. Every maximum M_3 consists of a maximum Γ-matching and edges saturating Y into $A \cup U_0$. Its Γ part may be replaced arbitrarily. If Y has degree two in J and avoids U_0, then

$$E(P_4) = E(J[U \cup Y]) \setminus M_\Gamma,$$

with A isolated. At every index one through four, the full restored total $N + \nu(P_i)$ is a certificate. The first three totals equal $c_1(F), c_2(F), c_3(F)$.

Proof. The booking assertion of Proposition 5.3 assigns all Y-vertices to distinct free odd components. Every vertex of an odd cycle or isolate is an exposed port. Thus arbitrary cubic owner edges can be added, with maximum internal row matchings avoiding the assigned ports. The result is a matching of size $p + \delta + h$, hence maximum by the row-plus-exterior upper bound.

The required source order inequality, with $z = a - p - \delta$, is

$$z \geq \epsilon + q_{\mathrm{def}} + s + 1 + 2v_6.$$

Together with $r = 3p + 2\delta + z$ and $|Y| \leq \beta + 4$, it gives $N \geq \beta + h - 1$. For clarity, put $s = 2t + 1$. Since $2v_6 \geq E - 3$ and $E = p + q_{\mathrm{def}} + \epsilon + t + 1$,

$$N - (\beta + h - 1) \geq p + 4\epsilon + q_{\mathrm{def}} - 2s + t - 8 + 2b.$$

For $\epsilon \geq 3$, use $s \leq \epsilon$, $p \geq 2$, $q_{\mathrm{def}} \geq 1$ to obtain a lower bound of one. For $\epsilon = 2$, $s = 1, t = 0$, and the same expression is $p + q_{\mathrm{def}} - 2 + 2b \geq 1$.

Lemma 6.3 now gives the first two paid ranks and clears the row. Each cubic has exactly three row neighbors, so these are all removed. Every other remaining edge meets Y or lies in Γ. Hence $\nu(P_3) \leq \mu + |Y| = h - s$. Restoration and $c_3(F) = r$ force equality. A maximum Γ-matching covers all of $U \setminus U_0$. The other $|Y|$ edges therefore join Y to $A \cup U_0$. Under the additional degree condition they join Y to A and consume its fourth and last row contact. The stated residual follows. The omission bound applies to all decreasing paid ranks, so Lemma 6.1 proves all certificates. □

A complete proof is recorded in [Rec]; the supporting development is collected in [II].

Theorem 6.14 (Degree six)**.** *The class $\mathcal{M}_6$ is empty.*

Proof. Use the actual selector of Proposition 6.12 in Proposition 6.13. Every hypothesis of the latter is then supplied by the original source. Its actual maximum Γ-matching leaves the cycle, g edges, and

isolates described by the selector proposition. The full certificate has total

$$\begin{aligned} N + \nu(P_4) &= u - |U| - |Y| + g + e_C - 1 \\ &= (r+s) - (s+\mu) - (\kappa + e_C) + g + e_C - 1 \\ &= r - L - \kappa - 1 = r - \beta - 1. \end{aligned}$$

The early row has $r = a + (p+\epsilon) + (p + q_{\mathrm{def}})$, so $r - \beta = a + p + \epsilon = c_4(F)$. The total is $c_4(F) - 1$, a contradiction. □

6.5. **Degree ten.**

Proposition 6.15 (Strict-sector selector and prefix)**.** *Let $F \in \mathcal{M}_{10}$ with $\delta > 0$. Choose any native row, any continuation, any maximum independent T_0 of the degree-two part of its terminal R, and any shortest whole odd cycle C of R. Retain every root and $Y = T_0 \cup V(C)$, and write $H = F[V(A) \cup X_{\mathrm{root}} \cup Y]$. Then*

$$|T_0| = \beta, \quad 2 \le e_C = (|C|+1)/2 \le 4, \quad |Y| = \beta + e_C, \quad h = s + \beta + e_C, \quad N = r - \beta - e_C.$$

The graph $R[Y]$ is C and isolates. Writing $q_7 = |B_7|$,

$$N \ge \max(0, \beta + h - 1), \qquad N \ge \max(0, h - q_7 - 2), \qquad N \ge e_C - 2.$$

For every missing pair (b_R, b_B), $h \le u - \max(b_R, b_B)$. Every three sequential maximum matchings have ranks $h + p + \delta, h + \beta, h + p$, saturate the retained exterior into A, and clear all row edges and all three A-contacts of every B_7 vertex. All retained exterior-internal edges survive.

A complete proof is recorded in [Rec]; the supporting development is collected in [II].

Proposition 6.16 (The degree-ten all-root lift)**.** *Let $F \in \mathcal{M}_{10}$ in either sector. Fix a native row and a continuation, and retain all roots and any subset Y of terminal degree-two vertices. Put $h = s + |Y|$, $N = r - |Y|$, $x = |B_7|$, $y = |B_6|$. Assume three edge-disjoint matchings each saturate the retained exterior into A and clear A, and $N \ge \max(0, h - x - 2)$. Then successive maximum matchings at indices three through six have ranks*

$$h - x, \quad h - x - y, \quad h - s + x + y, \quad h - s + x.$$

After them the entire seventh residual has form

$$E(P_7) = E(R[Y]) \,\dot\cup\, \{v\phi(v) : v \in Y\},$$

where ϕ is injective into $V(A) \cup X_{\mathrm{root}}$. The unique maximum matching is the displayed boundary matching. Deleting it leaves exactly $E(R[Y])$. If the initial triple is sequentially maximum with ranks $h + p + \delta, h + \beta, h + p$, the complete eight-entry rank table is

$$(h + p + \delta, h + \beta, h + p, h - x, h - x - y, h - s + x + y, h - s + x, h - s).$$

The restored ranks at indices one through seven equal the source costs.

Proof. Use the independent sides

$$I_i = V(A) \cup \bigcup_{10 - i \le j \le 7} B_j \qquad (3 \le i \le 7).$$

Their complementary cover sizes are the last five entries of the table. Proposition 5.5 gives $N + |C_i| = c_i(F)$. The first triple clears I_3. The deficient-rank contradiction used in Proposition 6.6 forces each subsequent cover rank to be exact. A vertex of the next phase has one remaining earlier contact, which is forced into the next matching. No edge of $R[Y]$ is removed.

At the seventh residual the earlier side is independent. Every Y-vertex had eight original earlier neighbors and seven of those edges have been removed. It has one last edge. The same deficient-rank argument at $c_7(F) = N + |Y|$ forces a matching saturating Y. Thus the last endpoints are distinct and the matching is unique. Its deletion leaves precisely $R[Y]$. The initial restored equalities follow from $N + h = u$. □

Proposition 6.17 (Degree-ten cycle absorption)**.** *For the strict-sector selector of Proposition 6.15, one may choose any whole odd cycle, without shortestness, for the following omission assertion:*

$$N = c_8(F) - e_C, \qquad N \geq 2(p-1) + 2\epsilon \geq 8.$$

For a shortest cycle, $N \geq (p-1)e_C + 2\epsilon$. Thus $N \geq e_C - 2$. If eight edge-disjoint matchings on this H have exact union $E(H) \setminus E(C)$, then every index-eight capacity and budget split at total

$$N + e_C - 1 = r - \beta - 1 = c_8(F) - 1$$

has a full-source certificate. Consequently such a matching family is impossible in a strict source.

For any integers $e_* \geq 2$, $N_* \geq e_* - 2$, the absorption data have finite simple maximum-degree-ten realizations outside the source class: attach eight private leaves to each vertex of a cycle of order $2e_* - 1$, add N_* disjoint edges $s_j t_j$, and omit the s_j. The eight corresponding spoke matchings are successive maximum matchings in any order and leave exactly the cycle.

Proof. Every unselected odd cycle of order $2w+1$ contributes $w+1 \geq 2$ omitted vertices. If C is shortest, this is at least e_C. Each positive even path of order $2l$ contributes $l + 1 \geq 2$ omitted vertices, since its internal degree-two path supplies $l - 1$ selected vertices. Summing proves the bounds. The full certificate is Lemma 6.1 with eight matchings and $g = 0$.

In the last construction every retained edge meets the original cycle vertex set, which is a cover of size $2e_* - 1$. Each remaining spoke matching has exactly that size. It is therefore maximum at every stage. The final certificate follows from the same restoration lemma. No minimal-source or cost-optimality assertion is made for this construction. □

Theorem 6.18 (The strict degree-ten sector)**.** *There is no $F \in \mathcal{M}_{10}$ with $c_2(F) < c_0(F)$.*

Proof. Proposition 6.15 supplies a compatible triple on one actual selected graph and every required margin. Proposition 6.16 extends it to eight successive maximum matchings leaving exactly the selected cycle. Proposition 6.17 supplies a certificate of total $c_8(F)-1$. This contradicts the definition of $c_8(F)$. □

Proposition 6.19 (The equality-sector prefix)**.** *Let $F \in \mathcal{M}_{10}$ satisfy $c_2(F) = c_0(F)$. Then $\delta = \epsilon = 0$. Put $z = a - p$. Every native row is $pC_3 + zK_1$, where $p \geq 5$, $z \geq 0$, and*

$$r = 3p + z = a + 2p, \qquad u = r + s.$$

Every native-row complement has word $6^t 5^{s-2t} 4^t$, with positive odd middle length. Its degree-seven and degree-three phases are empty. The exact costs are

$$(c_0, \ldots, c_9) = (u+p, u+p, u+p, u, u-t, r+t, r, r, r-p, a).$$

Fix any continuation with terminal R. Choose one vertex from each terminal triangle and complete any one triangle C. For the resulting Y, retain all roots and put $h = s + p + 2$, $N = r - p - 2$. With $v_{10} = \lfloor (2E-3)/8 \rfloor \geq 2$, every missing pair satisfies

$$u - b_R \geq s + p + v_{10} + 1 \geq h + 1, \qquad N - (p + h - 1) \geq 2v_{10} - 2 \geq 2.$$

There is a matching of size $p + h$ internally maximum on the row and assigning all retained exterior vertices to distinct exposed odd ports. Every three successive maximum matchings have rank $p + h$ and clear all row edges while preserving every exterior-internal edge.

A complete proof is recorded in [Rec]; the supporting development is collected in [II]. The missing-budget estimates are $b_R \leq k - E - v_{10}$, $k \geq 3E - 1 + 2v_{10}$. The prefix after that initial matching follows from Lemma 6.3.

Proposition 6.20 (The equality-sector terminal matching)**.** *For every graph and selector in Proposition 6.19, any seven successive maximum matchings leave*

$$E(P_7) = E(C) \,\dot\cup\, \{y\phi(y) : y \in Y\},$$

where ϕ is injective into the earlier side. The unique maximum matching is $\{y\phi(y) : y \in Y\}$, of size $p + 2$. After its deletion the entire remaining edge set is $E(C)$.

Proof. Apply Proposition 6.16 with $x = |B_7| = 0$, $y = |B_6| = t$, and the prefix and margin just supplied. Its empty phase sets are allowed. Its last matching is unique and leaves $R[Y]$, which is the chosen triangle and isolates. □

Theorem 6.21 (The equality degree-ten sector)**.** *There is no $F \in \mathcal{M}_{10}$ with $c_2(F) = c_0(F)$.*

Proof. The equality selector omits

$$N = r - p - 2 = 2p + z - 2 = a + p - 2 \geq 8.$$

Proposition 6.20 gives eight matchings leaving one triangle. For any capacity split, group those matchings between the degree parts. The triangle has matching number one; the N omitted vertices supply N indexed stars. At total $N + 1$, put the triangle and $x - 1$ stars in the first paid part when $x \geq 1$, and put everything in the second paid part when $x = 0$. This certifies every allocation and includes each omitted-incident edge once. Hence

$$c_8(F) \leq N + 1 = a + p - 1 < c_8(F) = a + p,$$

a contradiction. □

Theorem 6.22 (Degree ten)**.** *The class $\mathcal{M}_{10}$ is empty.*

Proof. Proposition 2.3 gives $c_2(F) \leq c_0(F)$ for every graph. Theorems 6.18 and 6.21 exclude the two exhaustive cases. □

6.6. **Degree four.**

Proposition 6.23 (The exact quartic cost interface)**.** *Let F be connected, four-regular, and factor-critical, with $\operatorname{gap}(F) \geq 1$ and $E = \operatorname{ex}(F) \geq 5$. Put $a = \alpha(F)$, $\sigma = 2E - 2 - (\nu(F) - a)$. For a maximal independent S, put*

$$H = F - S, \quad v = |V(H)|, \quad b = \nu(H) - c_1(H), \quad m = E - 1 - \sigma + b,$$

$$q_4 = m - b, \qquad \beta_4 = \nu(H) - 2m, \qquad \beta_4' = \nu(H) - b - m, \qquad \rho_4 = v - m.$$

These quantities are independent of S, and F is well-covered. They satisfy

$$q_4 = E - 1 - \sigma \geq \lceil (E-1)/4 \rceil \geq 1, \qquad \beta_4 = E + \sigma - 2b \geq E - \sigma \geq \lceil (E+3)/4 \rceil \geq 1,$$

$$\beta_4' = \beta_4 + q_4, \qquad v = a + \nu(H) + q_4.$$

The exact source costs are

$$c_0(F) = a + \nu(H) = v - q_4, \qquad c_1(F) = a + \nu(H) - b = \rho_4, \qquad c_2(F) = a + 2m - b = \rho_4 - \beta_4.$$

In particular,

$$c_2(F) = v - \nu(H) + b + q_4,$$

so omitting the positive correction q_4 gives an incorrect second cost.

A saturated frame is a pair (S, U) with U an independent m-set of degree-three vertices of H. For its terminal $R = F - (S \cup U)$,

$$c_1(R) = \beta_4', \quad c_2(R) = q_4, \quad \beta_4 = c_1(R) - c_2(R).$$

There are exactly q_4 whole odd terminal cycles.

For any $W \subseteq V(H)$, put $h = |W|$ and $N = v - h$. Then

$$c_0(F) - N = h - q_4, \quad c_1(F) - N = h - m, \quad c_2(F) - N = h - m - \beta_4.$$

If $h = m + \beta_4 + g + e$, then $N + g + e - 1 = c_2(F) - 1$. This last identity alone is not a certificate.

The quartic normalization input is $\nu(H) = 3E - 2 - \sigma$, $c_1(H) = \nu(H) - b$, $c_2(H) = 2m - b$, $v - \nu(H) = a + m - b$, $0 \leq b \leq \sigma \leq \lfloor 3(E-1)/4 \rfloor$, $m \geq \lceil (E-1)/4 \rceil + b$, and the exact independent-cut identities $c_i(F) = a + c_i(H)$ through index three with $\operatorname{ex}(H) = 0$. A complete proof is recorded in [Rec]; the supporting development is collected in [II].

The cost calculation. The stated normalization gives $q_4 = m - b = E - 1 - \sigma$ and $v = a + \nu(H) + q_4$. Substitution yields $\beta_4 = E + \sigma - 2b$ and $\beta_4' - \beta_4 = q_4$. The bounds on σ, b give the positive lower bounds. Also

$$(v - m) - (\nu(H) - 2m) = a + 2m - b = c_2(F).$$

The saturated-terminal identities are $\nu(R) = \nu(H) - m$, $c_1(R) = \nu(R) - b$, $c_2(R) = m - b$. They give the displayed terminal costs. Subtracting $N = v - h$ proves the final three identities and the debit formula. □

Proposition 6.24 (Saturation and the terminal alternative)**.** *Every $F \in \mathcal{M}_4$ has $E \geq 14$ and a saturated frame. For the wider source class in Proposition 6.23, assume a saturated frame is supplied. Every saturated terminal has exactly $q_4 = E - 1 - \sigma \geq 1$ odd cycles, all pentagons or heptagons. If $q_7(R)$ is its number of heptagons, then*

$$2q_4 + q_7(R) \leq 2(\sigma - b) + 1.$$

Consequently

$$E \leq 2\sigma - b + 1, \qquad 3b \leq 2\sigma.$$

The earlier $E = 8$ specialization, when considered under the corresponding minimal-source restrictions, forces $\sigma = 5$ and $0 \leq b \leq 3$.

A complete proof is recorded in [Rec]; the supporting development is collected in [II].

The terminal inequality from its stated inputs. Let $q_5(R)$ count pentagons and put $L(R) = q_5(R) + 2q_7(R) = q_4 + q_7(R)$. The required reflected-terminal inequality is

$$L(R) + q_5(R_D) + q_7(R_D) \leq 2(\sigma - b) + 1$$

for an actual reflected saturated terminal R_D. That terminal also has exactly q_4 odd cycles, giving the displayed inequality. Integrality implies $q_4 \leq \sigma - b$. Since $q_4 = E - 1 - \sigma$, this gives the bound on E. Combining it with $4\sigma \leq 3(E - 1)$ gives $3b \leq 2\sigma$. At $E = 8$, the stated earlier specialization gives $3b \leq 10$. □

Proposition 6.25 (Movable quartic selectors)**.** *Use a supplied saturated frame from Proposition 6.23. Put $W_{\mathrm{fr}} = S \cup U$, $R = F - W_{\mathrm{fr}}$. Its induced frame is $mK_2 + (a - m)K_1$. Let P_{fr} and Z be its endpoints and isolates, and let L_{fr} be its edge set. For $X \subseteq \{v \in V(R) : d_R(v) = 2\}$, put*

$$J_X = R[X], \quad K_X = F[W_{\mathrm{fr}} \cup X], \quad N_X = v - m - |X|, \quad \mathrm{gain}_X = |X| - \nu(J_X) - \beta_4.$$

Let G_X contain exactly the frame edges and all X–W_{fr} edges, and let B_X be the bipartite graph of the actual X–Z edges.

There are two edge-disjoint matchings with $L_{\mathrm{fr}} \subseteq M_0$ and $M_0 \cup M_1 = E(G_X)$ if and only if

$$|N_F(p) \cap X| \leq 1 \quad (p \in P_{\mathrm{fr}}), \qquad |N_F(z) \cap X| \leq 2 \quad (z \in Z),$$

and B_X has a matching saturating X. Every such pair is sequentially maximum, with ranks $m + |X|, |X|$, and leaves exactly $E(J_X)$. Moreover

$$N_X \geq \nu(H) + q_4 \geq \nu(J_X) + 1, \quad N_X + |X| = c_1(F), \quad N_X + \nu(J_X) = c_2(F) - \mathrm{gain}_X\,.$$

Every index-two allocation at the last total has a full-source certificate. Therefore $\mathrm{gain}_X \leq 0$.

Under only the two load bounds, the inclusion-minimal deficient X-subsets are exactly the X-sides of whole B_X paths with both endpoints in X, including isolated X-vertices. Let there be d_X such paths. Deleting one arbitrarily chosen X-vertex from each repairs all ports. For the resulting set of deleted vertices A,

$$\mathrm{gain}_X \leq \eta_{J_X}(A) := |A| + \nu(J_X - A) - \nu(J_X) \leq d_X.$$

If $d_X = 1$ and $\mathrm{gain}_X \geq 1$, then $\mathrm{gain}_X = 1$ and every possible deleted vertex is exposed by a maximum J_X-matching. Such a vertex lies on an odd cycle or in an odd-numbered position on an odd-order path, with an isolate included.

A deficient path has exactly two edges to P_{fr}, at its X-endpoints, with distinct frame endpoints. If these endpoints are partners, it closes with their frame edge to a whole odd component of G_X. Otherwise it joins two distinct frame edges.

Finally, choose a canonical independent D in R by taking maximum independent sets on odd cycles, centers on P_3, opposite pairs on C_4, internal alternating sets on even paths, and no isolates. Deleting one D-vertex from each reflected odd cycle gives a neutral D_0 with

$$|D_0| = \beta_4, \quad R[D_0] \text{ edgeless}, \quad N_{D_0} = c_2(F).$$

For arbitrary releases $Z_- \subseteq D_0$ and additions $Z_+ \subseteq D_2(R) \setminus D_0$, set $X = (D_0 \setminus Z_-) \cup Z_+$. Then

$$N_X = c_2(F) + |Z_-| - |Z_+|, \qquad \operatorname{gain}_X = |Z_+| - |Z_-| - \nu(J_X).$$

If $J_X = C_{2e-1} \sqcup B$ and $g = \nu(B)$, the proposed unit saving $|Z_+| - |Z_-| = g + e$ forces a load violation or a deficient port path.

The frame inputs include the induced matching structure of every saturated frame, the bound $m \le \sigma$, and the canonical-reflection theorem giving the neutral selector D_0. A complete proof is recorded in [Rec]; the supporting development is collected in [II]. The matching criterion and certificate deductions below are proved from those precise frame inputs.

Proof. Every selected vertex has two distinct frame neighbors by four-regularity. If two matchings partition G_X with all frame edges in M_0, its degrees are at most two. This gives the two load bounds. Every X-vertex needs one edge of M_0, whose frame endpoint must be in Z, since the endpoints are already occupied by frame edges. Thus B_X saturates X.

Under the load bounds, B_X has maximum degree two. Its components are paths, even cycles, and isolates. The only deficient components are paths with both endpoints in X. Their entire X-side has one more vertex than its neighborhood. Deleting any one X-vertex splits such a path into two balanced paths. Alternating edges saturate their X-vertices. Every other component saturates its whole X-side; the matching may also cover all degree-two Z-vertices. This proves both the criterion and the assertion about inclusion-minimal deficient subsets.

Choose such a matching L in B_X and put $M_0 = L_{\mathrm{fr}} \cup L$. It covers every degree-two vertex of G_X, so the entire complement $M_1 = E(G_X) \setminus M_0$ is a matching. Their sizes are $m + |X|$ and $|X|$. The sets $U \cup X$ and X are vertex covers before and after M_0, respectively, proving sequential maximality for every admissible pair.

Port injectivity gives $|X| \le |Z| = a - m$. Hence $N_X \ge v - a = \nu(H) + q_4 \ge \nu(J_X) + 1$. Lemma 6.1 gives the complete certificate. Proposition 6.23 gives its exact cost comparison.

For a simultaneous repair set A, the new omitted count is $N_X + d_X$, so its certified total is

$$N_X + d_X + \nu(J_X - A) = c_2(F) - \operatorname{gain}_X + \eta_{J_X}(A).$$

Deleting $|A|$ vertices lowers a matching number by between zero and $|A|$, proving $0 \le \eta_{J_X}(A) \le d_X$. The exact source cost gives the lower inequality for every repair. The one-path assertion follows by integrality. On a path of odd order, deleting an odd-position vertex preserves the matching number and deleting an even-position vertex lowers it by one. Even paths and even cycles lose one under every vertex deletion; odd cycles lose none. This proves the exposure description.

An internal X-vertex on a deficient port path has both frame neighbors in Z. Each endpoint has one remaining neighbor in P_{fr}; an isolated X-vertex has two. The load bound and simplicity make the two frame endpoints distinct. This proves the physical boundary assertion.

The canonical-reflection input supplies D_0. All its vertices have degree two in R. Counting additions and releases gives the last two identities. Matching additivity gives $\nu(C_{2e-1} \sqcup B) = e - 1 + g$. A proposed unit saving has gain one, so the earlier certificate criterion forces an obstruction. □

Proposition 6.26 (Exclusion of saturated pentagons)**.** *No supplied saturated frame of a connected four-regular factor-critical graph with positive gap and excess at least five has a whole pentagon terminal component.*

Proof. Suppose the terminal contains a whole pentagon C. Fix a canonical D and keep all choices outside C fixed. The canonical-reflection input says that $R_D = F[W_{\mathrm{fr}} \cup D]$ is an actual saturated terminal. It has permitted components $K_1, P_2, P_3, P_4, C_3, C_4, C_5, C_7$. Every designated odd cycle or positive even path contains exactly one original frame edge. The original D–Z graph has exactly q_4 deficient D-ended paths, corresponding to reflected odd cycles.

The required pentagon parent theorem says that the two frame-parent sets of its vertices are equal or disjoint. An equality class has one vertex or two nonadjacent vertices. At most one edge joins the pentagon to P_{fr}; its contacted vertex is a singleton class. Its other parent is in Z and meets no other pentagon vertex.

A canonical pentagon choice is, after rotation, $\{0,2\}$ on $0,1,2,3,4$. The residual edge $3\,4$ is a frame edge in the reflected frame. Both selected vertices have a neighbor at one of its endpoints. Consider a selected vertex x with both old parents in Z. Its reflected component is neither K_1 nor P_2. It cannot be a designated P_4, since the two old frame endpoints, two Z-parents, and x would require five vertices. A designated C_3 or C_5 has too few old frame vertices to contain both parents and both endpoints. A reflected C_7 is excluded by the required theorem that a saturated heptagon has no frame-endpoint neighbor. Thus x lies in a whole P_3 centered at x or a whole C_4. For the possible contacted vertex with parents $\{p,z\}$, the required contact theorem gives the whole reflected path $p'-p-x-z$. No selected pentagon vertex therefore lies on a reflected odd cycle.

If an outside selected vertex shares a Z-parent of a pentagon vertex, choose a canonical pentagon pair containing that vertex. The reflected whole P_3 alternative is impossible at a shared parent. The whole C_4 alternative forces equality of both parent pairs. For a parent pair $Q \subseteq Z$, let k_Q count its pentagon vertices and l_Q its fixed outside selected vertices. Its whole pentagon class can be included in a canonical pair. The reflected degree bound then gives $k_Q + l_Q \leq 2$. The possible contacted pair has no outside selected neighbor.

Now put $X = D \cup V(C)$. Each affected two-parent component of its port graph is $K_{k_Q+l_Q,2}$ with left size one or two. It is a right-ended path or a balanced square, and saturates its left side. The contacted component, when present, is one balanced edge. Outside these components the port graph is unchanged. Thus its deficient paths are precisely the original q_4 paths, none meeting C. The endpoint loads are at most one and the isolate loads at most two.

Delete any one selected vertex from each deficient path. The resulting D_0 has size β_4, retains the two selected pentagon vertices, and $Y = D_0 \cup V(C)$ has size $\beta_4 + 3$. Proposition 6.25 repairs all ports and gives two sequential maximum matchings of ranks $m+\beta_4+3, \beta_4+3$. Their entire remainder is the pentagon and isolates. The omitted count is $N = c_2(F) - 3 \geq 3$. Restoration certifies every index-two allocation at $N+2 = c_2(F) - 1$, a contradiction. □

A complete proof is recorded in [Rec]; the supporting development is collected in [II].

Proposition 6.27 (Exclusion of saturated heptagons)**.** *No supplied saturated frame of a connected four-regular factor-critical graph with positive gap and excess at least five has a whole heptagon terminal component.*

Proof. Suppose C is a whole terminal heptagon. The required parent theorem gives two parents in Z for each vertex, equal or disjoint parent pairs, and equality classes of size one or two nonadjacent vertices. No frame endpoint has a neighbor on C.

Fix outside canonical choices. For any prescribed $x \in C$, rotate the cycle so that $x = 0$ and choose $\{0,2,4\}$ on it. The unselected vertices induce the edge $5\,6$, which is a frame edge in that actual reflection, and x meets its endpoint 6. The component count used in the pentagon proof excludes a reflected P_4, C_3, C_5 containing x with its two old isolate parents. The frame-contact prohibition excludes a reflected C_7. Thus x lies in a whole P_3 centered at x or a whole C_4.

If an outside canonical vertex shares one parent, this reflection forces a whole C_4 and equality of its parent pair with that of x. For each pair Q, let k_Q count its heptagon vertices and l_Q its outside canonical vertices. The full class on C can be included in a maximum independent triple. For a two-vertex class, the two nonadjacent vertices have closed neighborhood union of size at most six, so a third vertex can be added. The reflected degree bound gives $k_Q + l_Q \leq 2$.

Consequently completing all of C changes its affected port components only to $K_{1,2}$ or $K_{2,2}$. They are nondeficient. Every old deficient path remains unchanged and avoids C. The endpoint load remains at most one and the isolate load at most two. Delete any one selected vertex from each of the q_4 old deficient paths. The repaired set Y has size $\beta_4 + 4$ and its induced terminal graph is exactly C and isolates.

Proposition 6.25 supplies two sequential maximum matchings of ranks $m+\beta_4+4, \beta_4+4$. Their entire remainder is the same heptagon, of matching number three. The omitted count is $N = c_2(F) - 4 \geq 4$.

The full restored total is $N+3 = c_2(F)-1$, a contradiction. The rotations used to test individual parent contacts kept all outside choices fixed; they did not identify matchings from different reflections. □

A complete proof is recorded in [Rec]; the supporting development is collected in [II].

Theorem 6.28 (Supplied saturated sources)**.** *There is no triple (F,S,U) in which F is connected, four-regular, factor-critical, has positive gap and excess at least five, and (S,U) is a saturated frame with the parameters of Proposition 6.23.*

Proof. Proposition 6.24 gives $q_4 = E-1-\sigma \geq 1$ whole odd terminal cycles, each a pentagon or a heptagon. Propositions 6.26 and 6.27 exclude both possibilities in the same supplied frame. Thus a terminal with no odd cycle cannot evade the contradiction: the positive identity for q_4 forbids it. □

Theorem 6.29 (Degree four)**.** *The class $\mathcal{M}_4$ is empty.*

Proof. An alleged $F \in \mathcal{M}_4$ is connected, four-regular, factor-critical, has gap one and excess at least fourteen by Proposition 5.1. Proposition 6.24 supplies an actual saturated frame, before any optimizing choice is made. This gives a triple forbidden by Theorem 6.28. No saturation assumption remains in the whole-degree conclusion. □

7. Proof of the main theorem

In this section we assemble the whole-degree exclusions into Theorem 1.1.

Proof of Theorem 1.1, subject to the deferred inputs. Theorems 6.29, 6.14, 6.11, 6.22, and 6.7 establish the five assertions $N_4, N_6, N_8, N_{10}, N_{12}$ on the identical classes of Definition 2.4. The degree-ten theorem covers both equality and strict sectors. The degree-four theorem discharges its frame-existence condition. Every class uses arbitrary proper subgraphs, not only induced subgraphs.

Apply Theorem 4.7 with $q = 0$. Every graph with fewer than $L = \sum_{k=2}^{s+t} \ell_k$ edges has a coloring avoiding the two original targets. Thus $\widehat{R}(F_1, F_2) \geq L$. Proposition 3.2 gives the reverse inequality and an attaining host.

Every diagonal is nonempty: choose $i = \max(1, k-t)$, $j = k-i$. The inequalities $1 \leq i \leq s$, $i \leq k-1$, and $i \geq k-t$ give $1 \leq j \leq t$. All diagonal values are positive integers. No step requires strict inequalities between successive component sizes, a parity condition, or a lower bound beyond one. Hence repeated sizes and unit stars are included. When $s = t = 1$, the sole diagonal gives $\widehat{R}(K_{1,n_1}, K_{1,m_1}) = n_1 + m_1 - 1$. Either target length may separately equal one. □

Remark 7.1 (Logical status of the assembly). The assembly leaves no additional source hypothesis once its technical inputs are established. The present manuscript still contains deferred proofs of those inputs. Accordingly the completed argument here is conditional on their validity; the unconditional claim of Theorem 1.1 requires completion and independent checking of those proofs.

8. Concluding remarks

The certificate construction isolates two obligations. An actual retained graph must first admit compatible sequential matchings, with each choice made on the residual of the preceding one. The final partition must then restore every original edge incident with an omitted vertex. Exact source costs force the intermediate matching ranks, while the odd-cycle remainder saves one unit in the final total.

The five degree exclusions suffice for the formula. No necessity of this particular reduction is claimed. The immediate remaining work for this manuscript is to supply the deferred technical proofs, reconcile the omitted assertions identified in the preservation flag, complete the reference ledger, and run the compilation and independent mathematical checks.

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